ECONOMICS

Complete Diagram Atlas

Micro · Macro · International · Development

Environmental · Labour · Public · Game Theory

60 fully-labelled, colour diagrams
with concept explanations

Equilibrium points · Shifts · Shaded areas

Exam-oriented visual reference for UPSC / BPSC / UGC-NET / Assistant Professor

Contents

1. Market and Price Theory

- Demand Curve
- Supply Curve
- Market Equilibrium
- Individual vs Market Demand
- Individual vs Market Supply

2. Consumer Theory

- Indifference Curve
- Budget Line
- Consumer Equilibrium
- Income Consumption Curve (ICC)
- Price Consumption Curve (PCC)
- Engel Curve
- Demand Curve Derived from PCC

3. Production Theory

- Total Product (TP) Curve
- Average & Marginal Product
- Isoquant Curve
- Isocost Line
- Expansion Path

4. Cost Theory

- Total Cost Curves (TC, TFC, TVC)
- Average Cost Curves (AFC, AVC, ATC)
- Marginal Cost & U-shaped Costs
- Long-Run Average Cost (LAC) — Envelope
- LAC and LMC Relationship

5. Revenue Theory

- Revenue under Perfect Competition
- Revenue under Imperfect Competition

6. Firm Equilibrium

- Perfect Competition Equilibrium
- Monopoly Equilibrium
- Monopolistic Competition (Long Run)
- Kinked Demand Curve (Oligopoly)

7. Welfare Economics

- Production Possibility Frontier (PPF)
- Contract Curve (Edgeworth Box)
- Utility Possibility Frontier (UPF)
- Social Welfare Function (Bergson-Samuelson)

8. National Income & Output

- Consumption & Saving Functions
- Investment Function

9. Keynesian Analysis

- Keynesian Cross (Income Determination)
- Effective Demand (AD–AS)
- Aggregate Demand & Supply (AD–AS Model)

10. Money & Banking

- Liquidity Preference (Money Demand)
- Money Market Equilibrium

11. IS-LM Model

- IS-LM Equilibrium
- Fiscal Expansion in IS-LM

12. Inflation & Employment

- Phillips Curve (Short Run)
- Expectations-Augmented / Long-Run Phillips Curve

13. Growth Theory

- Solow Growth Model

14. International Economics

- Offer Curve (Reciprocal Demand)
- Terms of Trade & Trade Indifference
- J-Curve Effect
- Tariff: Welfare Effects

15. Development Economics

- Lorenz Curve
- Kuznets Curve
- Lewis Model (Surplus Labour)

16. Environmental Economics

- Externality: Private vs Social Cost
- Marginal Social Benefit vs Cost (Optimal Pollution)
- Environmental Kuznets Curve

17. Labor Economics

- Labour Demand & Supply
- Backward-Bending Labour Supply

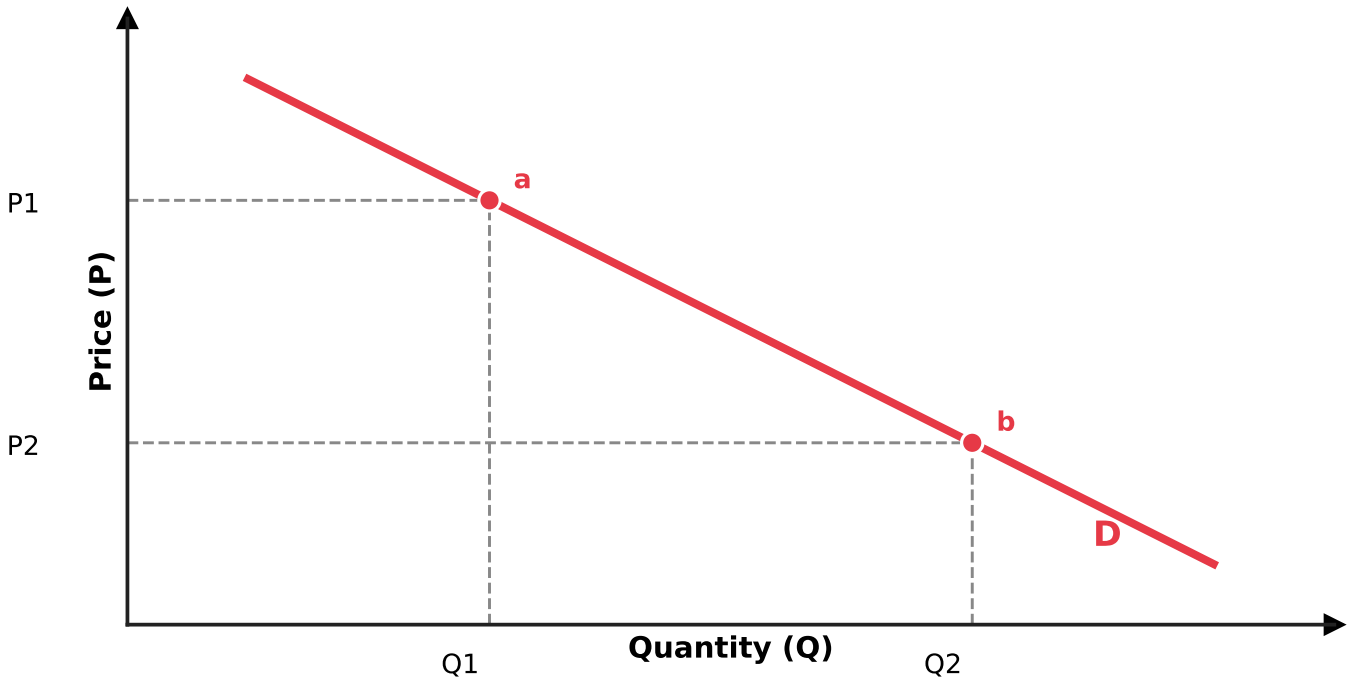
18. Public Economics

- Laffer Curve
- Tax Incidence (Burden Sharing)

19. Game Theory & Industrial Economics

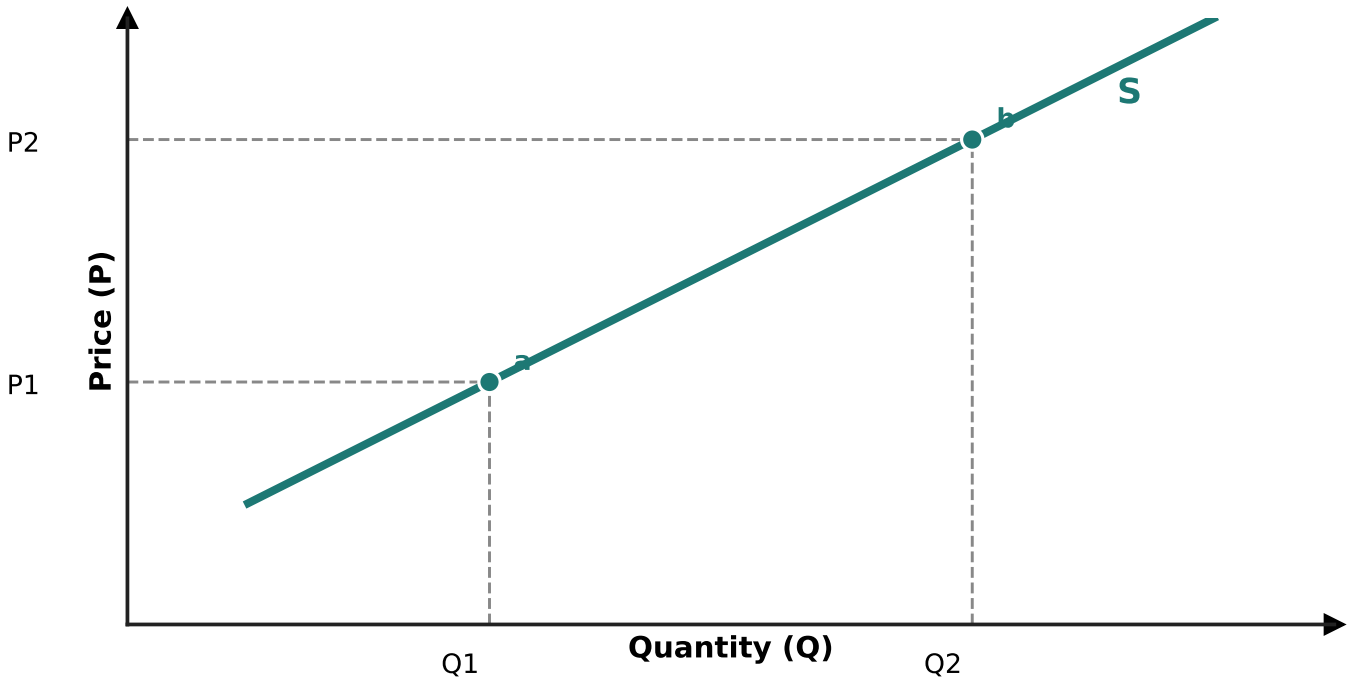
- Cournot Reaction Curves
- Learning / Experience Curve

1. Demand Curve



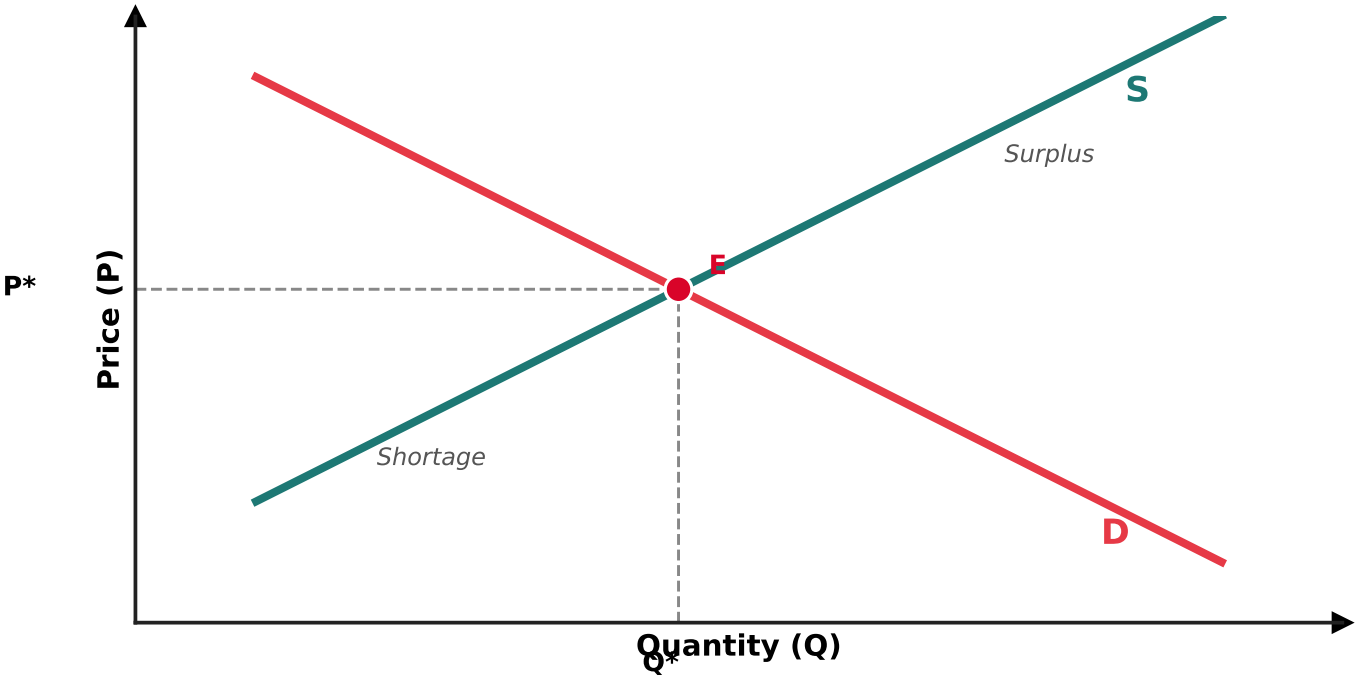
Shows the inverse relationship between price and quantity demanded (Law of Demand). As price falls from P1 to P2, quantity demanded rises from Q1 to Q2, giving the curve its downward slope.

2. Supply Curve



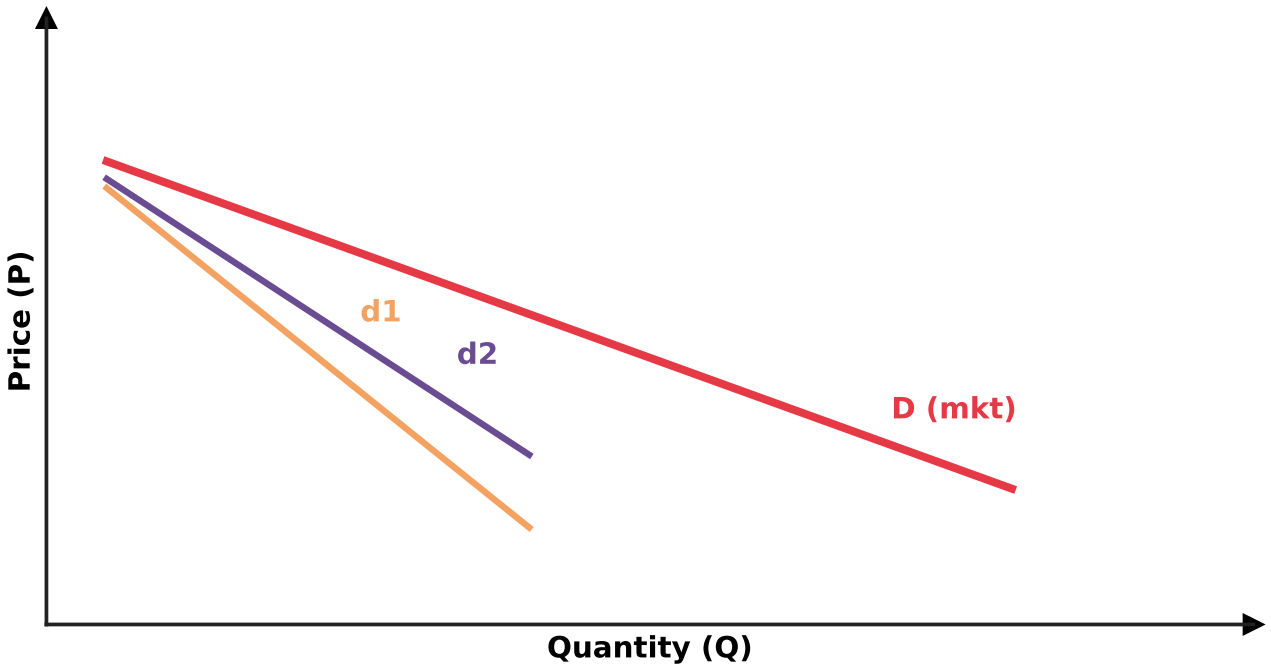
Shows the direct relationship between price and quantity supplied (Law of Supply). Higher prices incentivise greater production, so the curve slopes upward from left to right.

3. Market Equilibrium



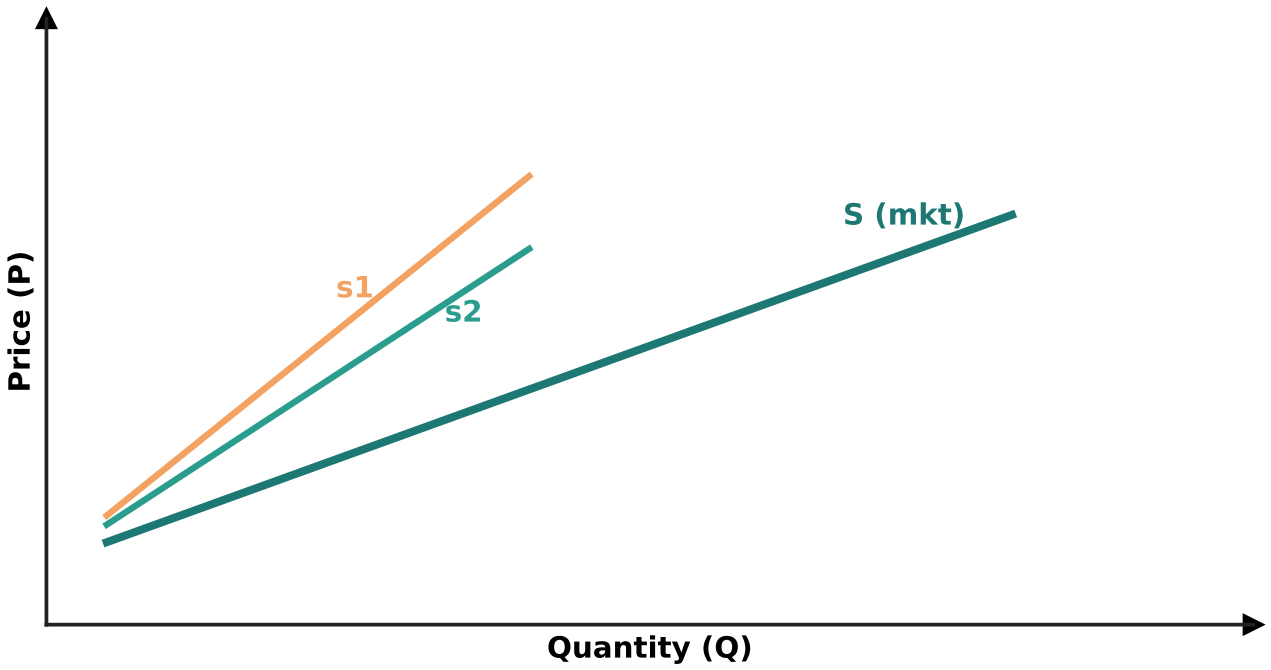
Equilibrium occurs where demand and supply intersect (E), determining equilibrium price P* and quantity Q*. Above P* there is surplus; below P* there is shortage, both pushing price back to equilibrium.

4. Individual vs Market Demand



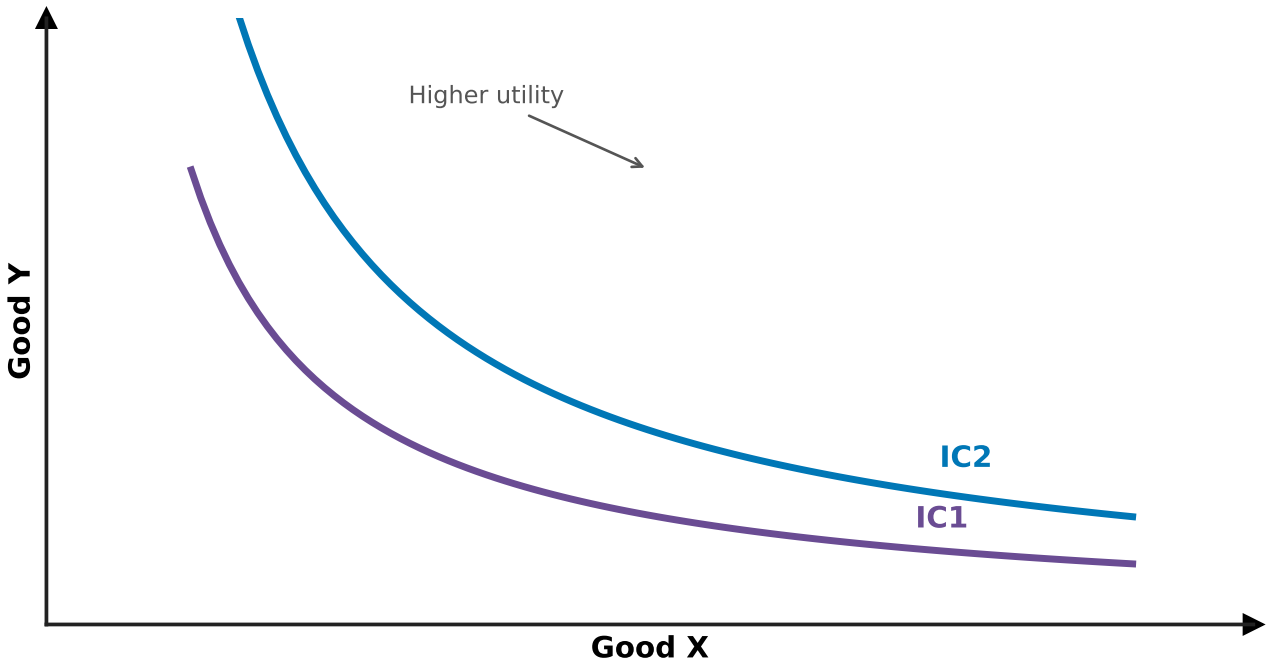
Market demand is the horizontal summation of all individual demand curves (d1 + d2 = D). At each price, individual quantities are added to obtain total market quantity demanded.

5. Individual vs Market Supply



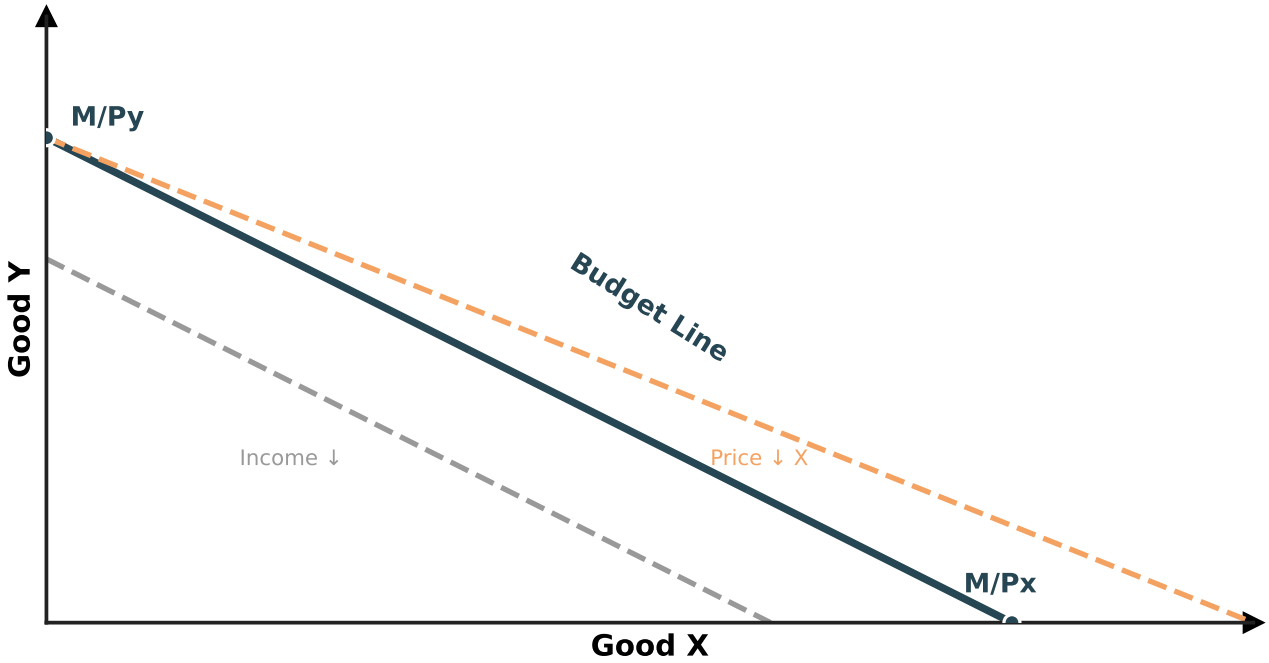
Market supply is the horizontal summation of individual firm supply curves (s1 + s2 = S). Total quantity supplied at each price equals the sum of what every firm offers.

6. Indifference Curve



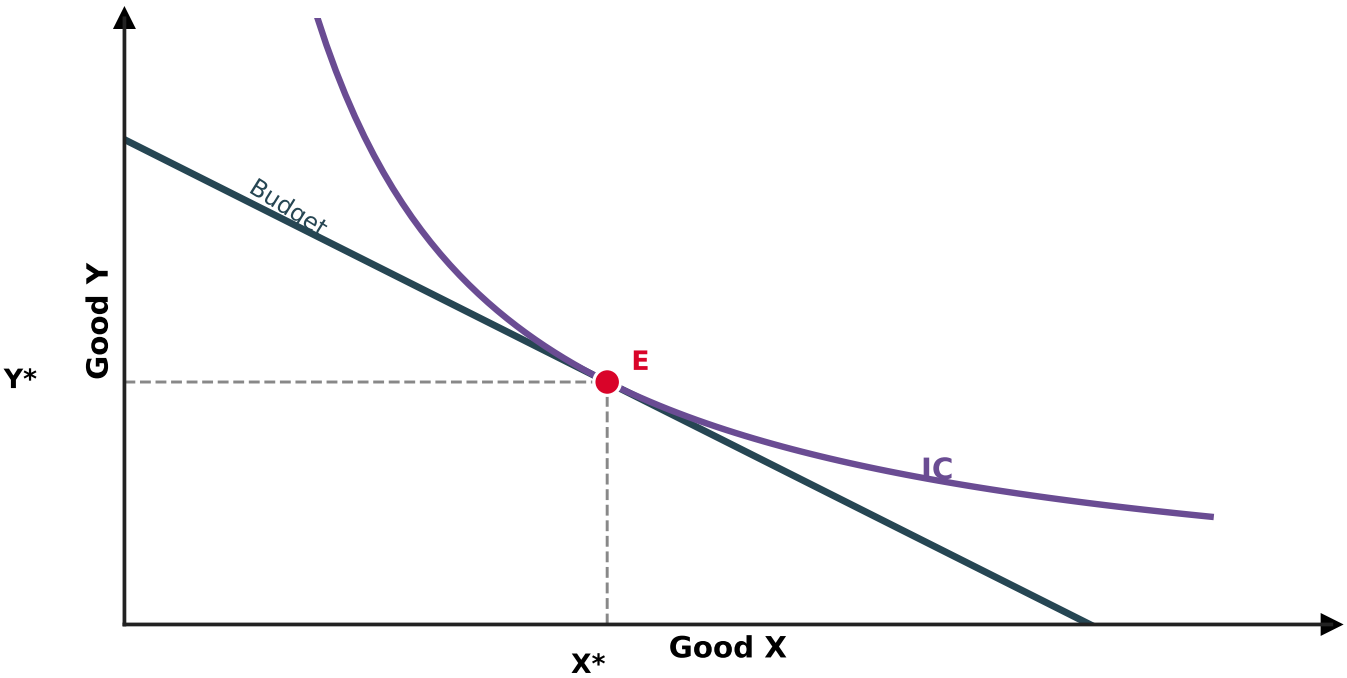
An indifference curve joins all consumption bundles giving equal satisfaction. It is convex to the origin due to diminishing MRS; higher curves (IC2) represent higher utility than lower ones (IC1).

7. Budget Line



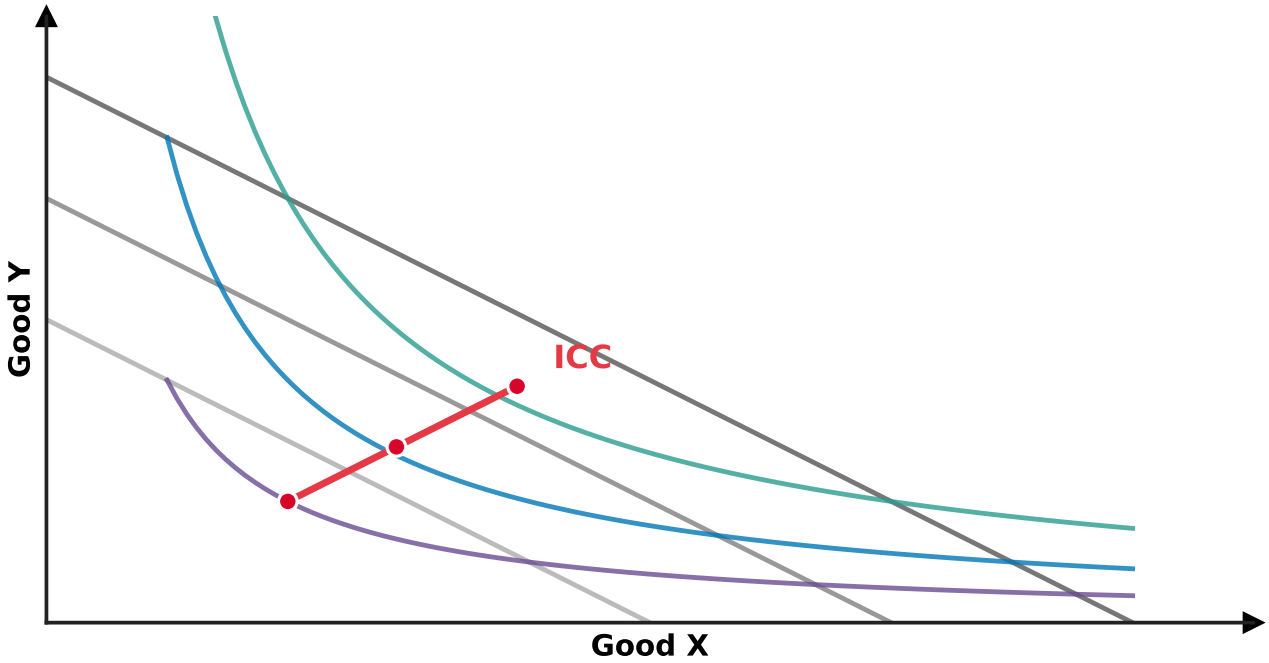
The budget line shows all combinations of two goods affordable at given prices and income (Px·X + Py·Y = M). Its slope equals the price ratio (−Px/Py); income or price changes shift or pivot the line.

8. Consumer Equilibrium



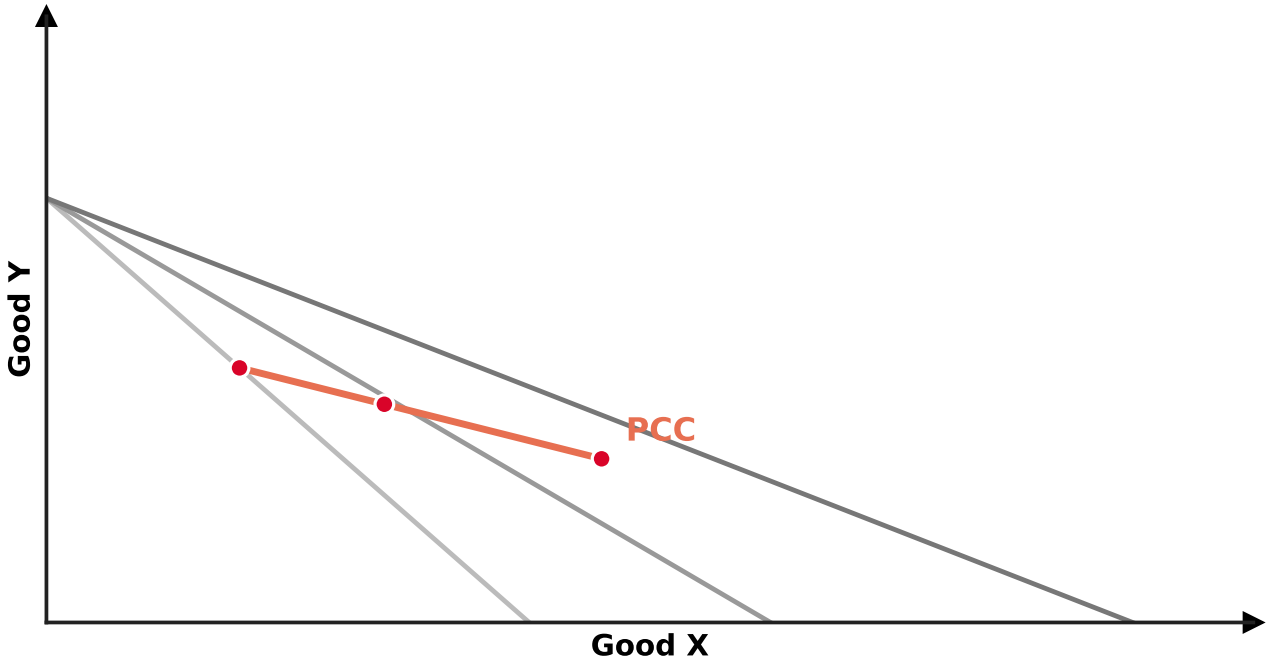
Consumer equilibrium is at the tangency (E) of the budget line and the highest attainable indifference curve, where MRS = Px/Py. The consumer maximises utility subject to the budget constraint.

9. Income Consumption Curve (ICC)



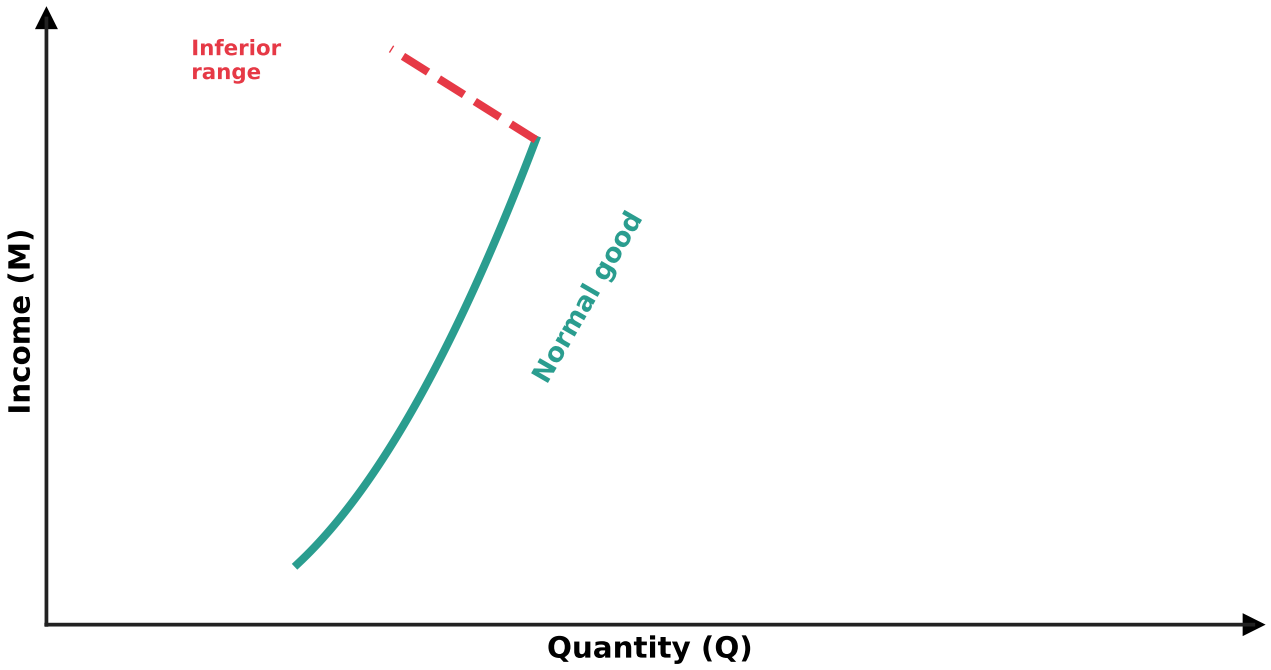
The ICC traces consumer equilibria as income rises (parallel budget-line shifts), holding prices constant. It shows how optimal consumption of both goods changes with income (upward for normal goods).

10. Price Consumption Curve (PCC)



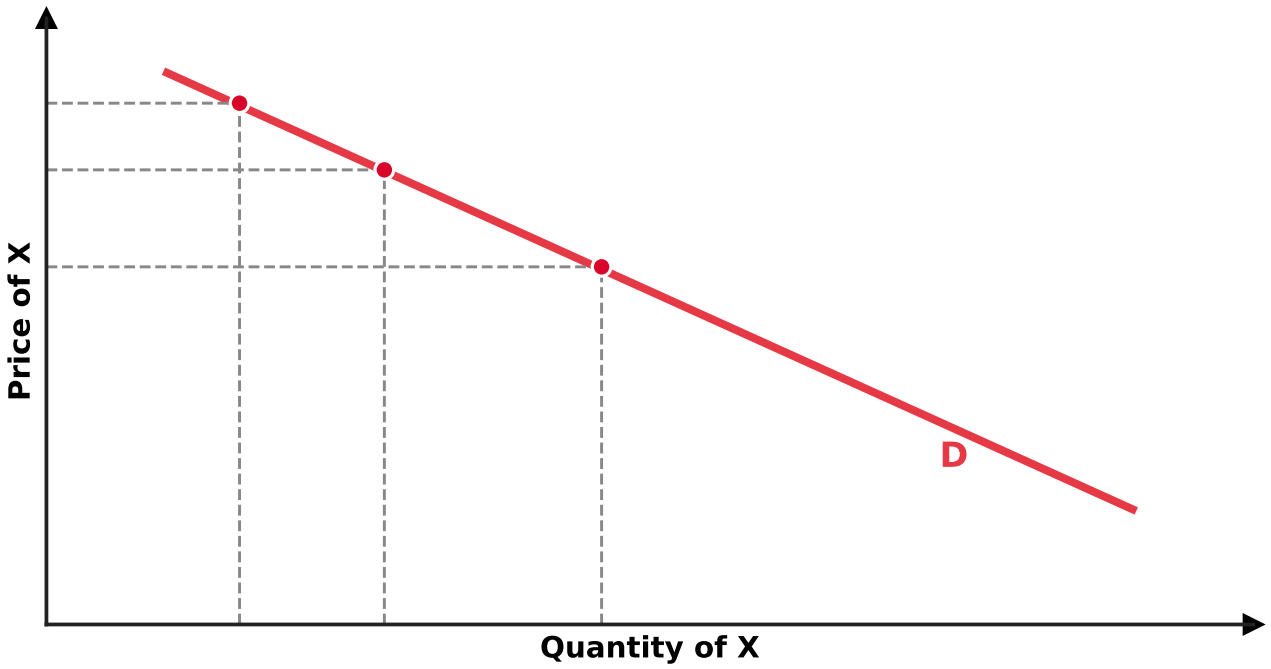
The PCC joins equilibrium points as the price of one good changes (budget line pivots), income constant. It is the basis for deriving the individual demand curve for that good.

11. Engel Curve



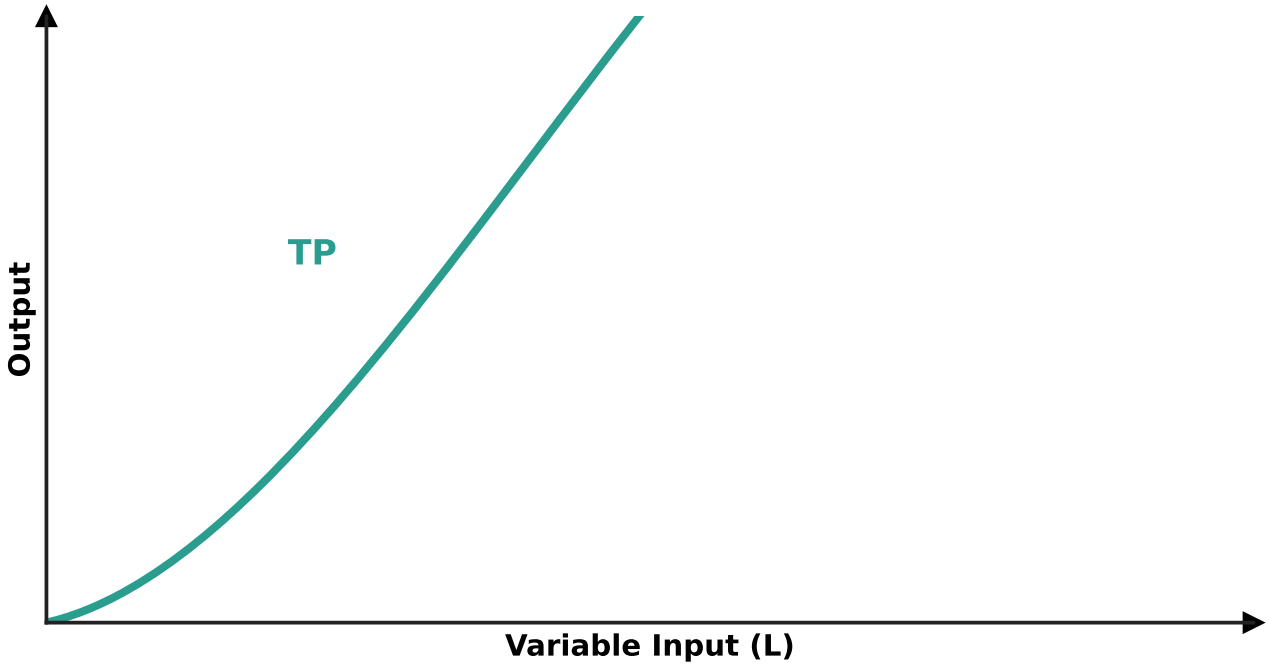
The Engel curve plots quantity demanded of a good against consumer income. For normal goods it slopes upward; for inferior goods it bends backward as income rises beyond a point.

12. Demand Curve Derived from PCC



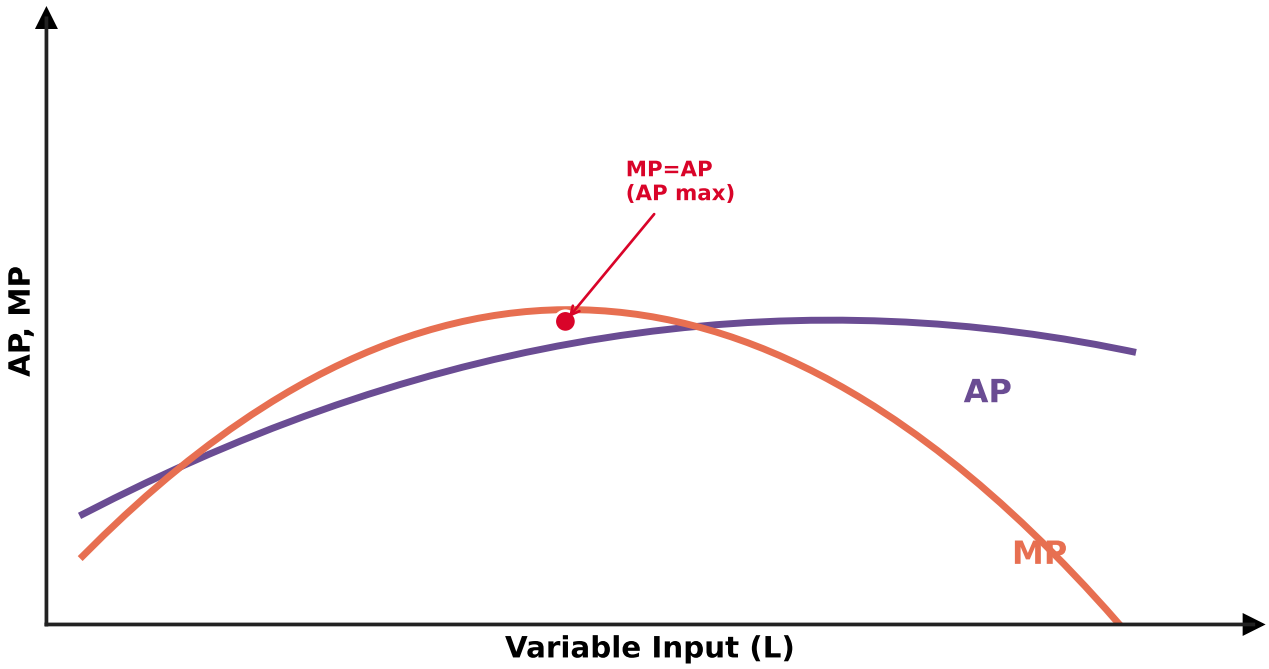
Mapping price–quantity pairs read off the PCC yields the downward-sloping individual demand curve. Each pivot of the budget line (a price change) corresponds to one point on the demand curve.

13. Total Product (TP) Curve



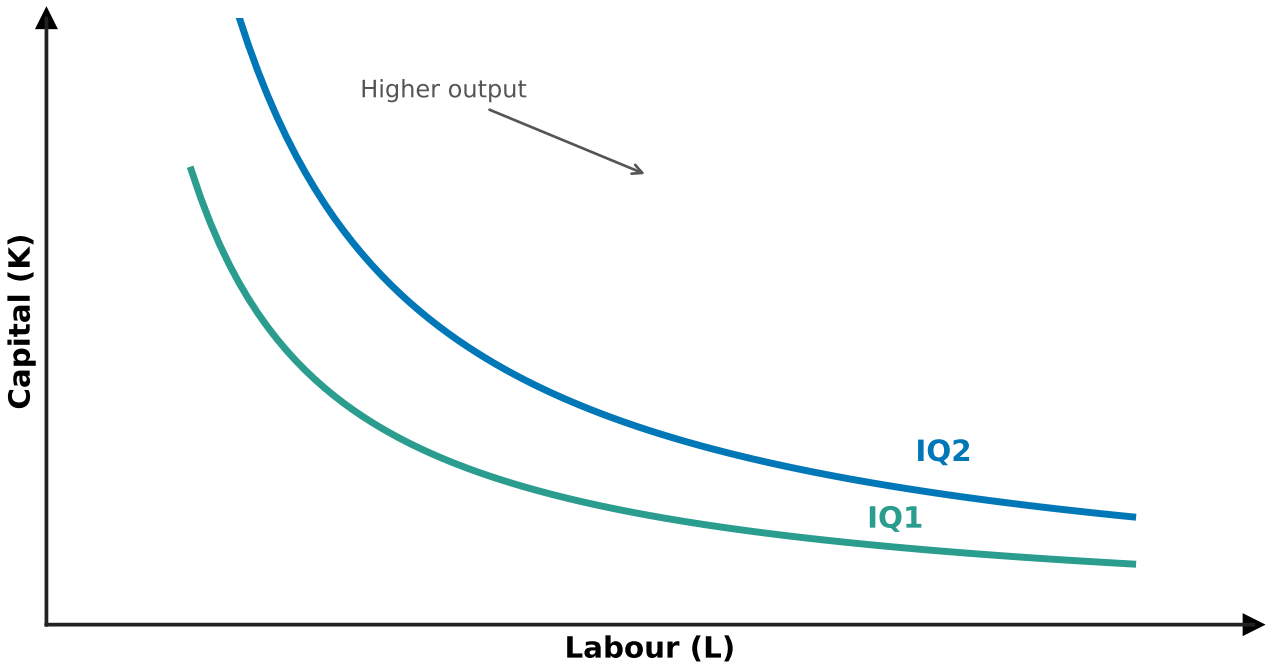
TP shows total output as the variable input increases. It rises at an increasing rate, then a diminishing rate, reaching a maximum (Law of Variable Proportions) before eventually declining.

14. Average & Marginal Product



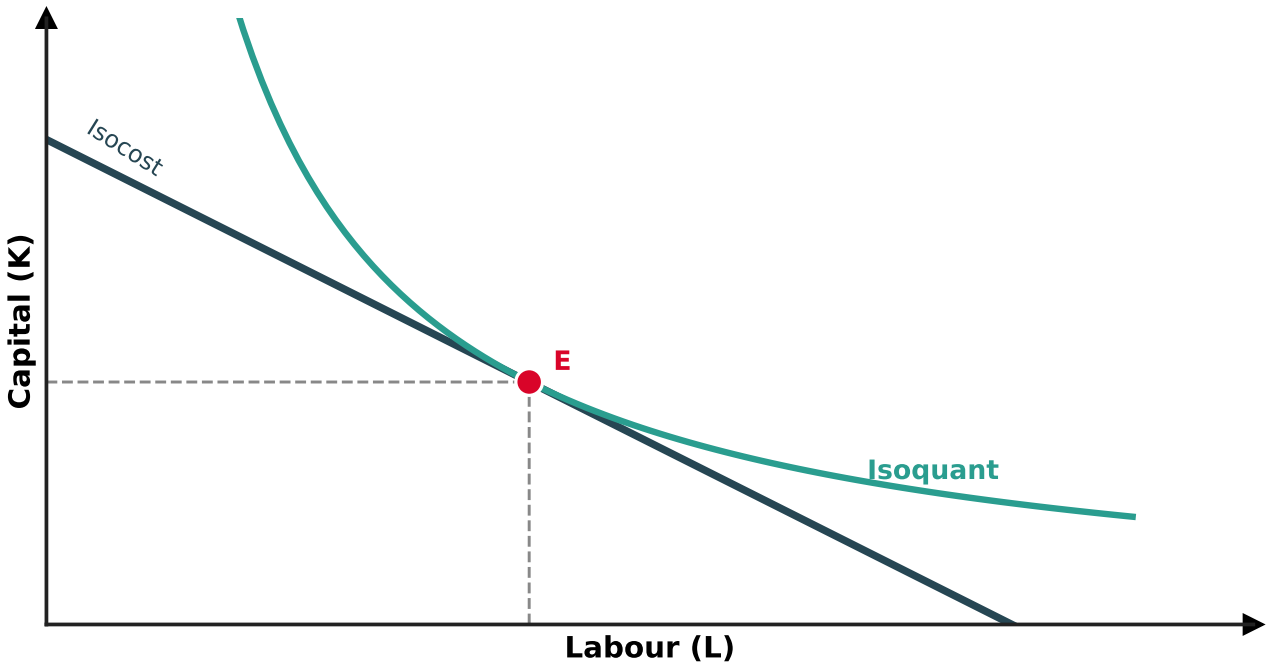
AP and MP are derived from TP. MP cuts AP at AP's maximum; the three stages of production are defined by their relationship. MP becomes zero when TP is maximum and negative thereafter.

15. Isoquant Curve



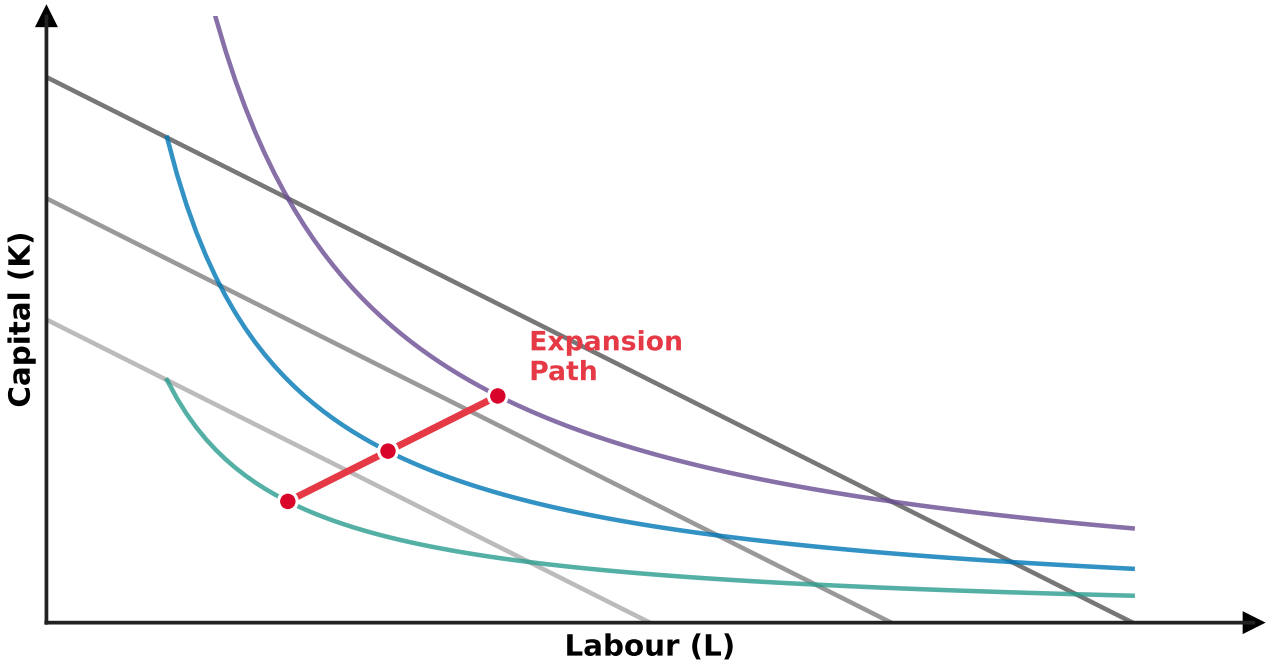
An isoquant shows all combinations of two inputs (L, K) yielding the same output. It is convex due to diminishing MRTS; higher isoquants represent higher output levels.

16. Isocost Line



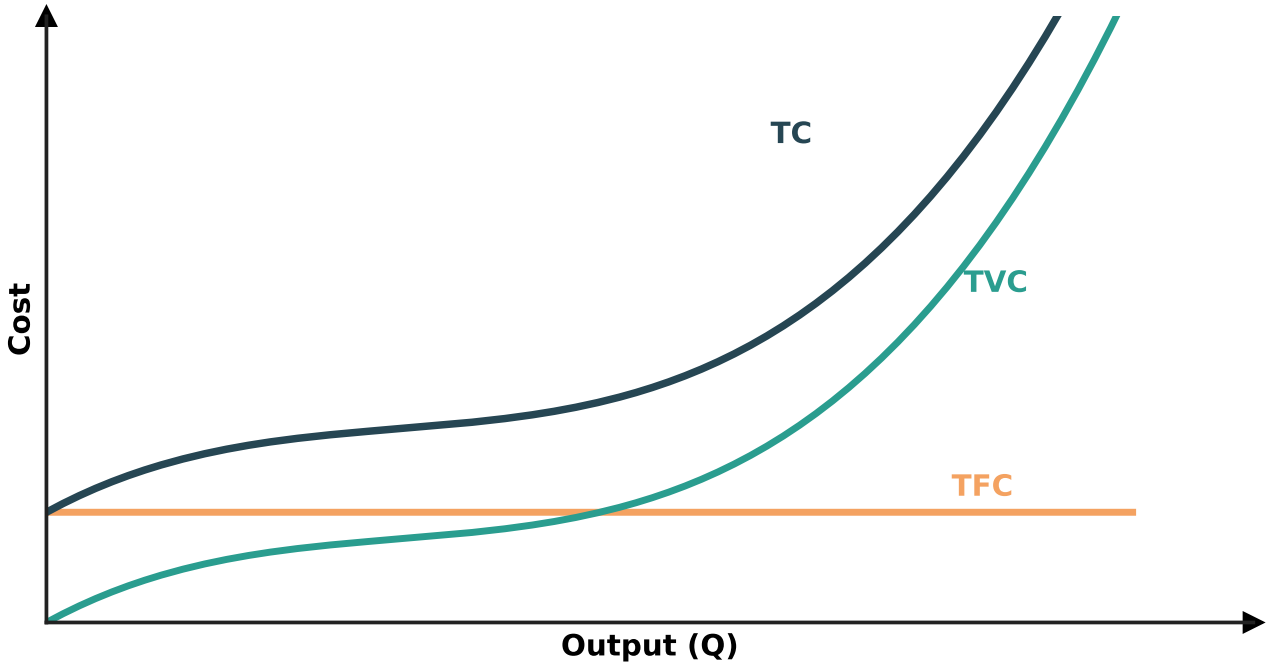
The isocost line shows input combinations costing the same total outlay (w·L + r·K = C). Its slope is the input price ratio (−w/r). Producer equilibrium is where an isoquant is tangent to the isocost line.

17. Expansion Path



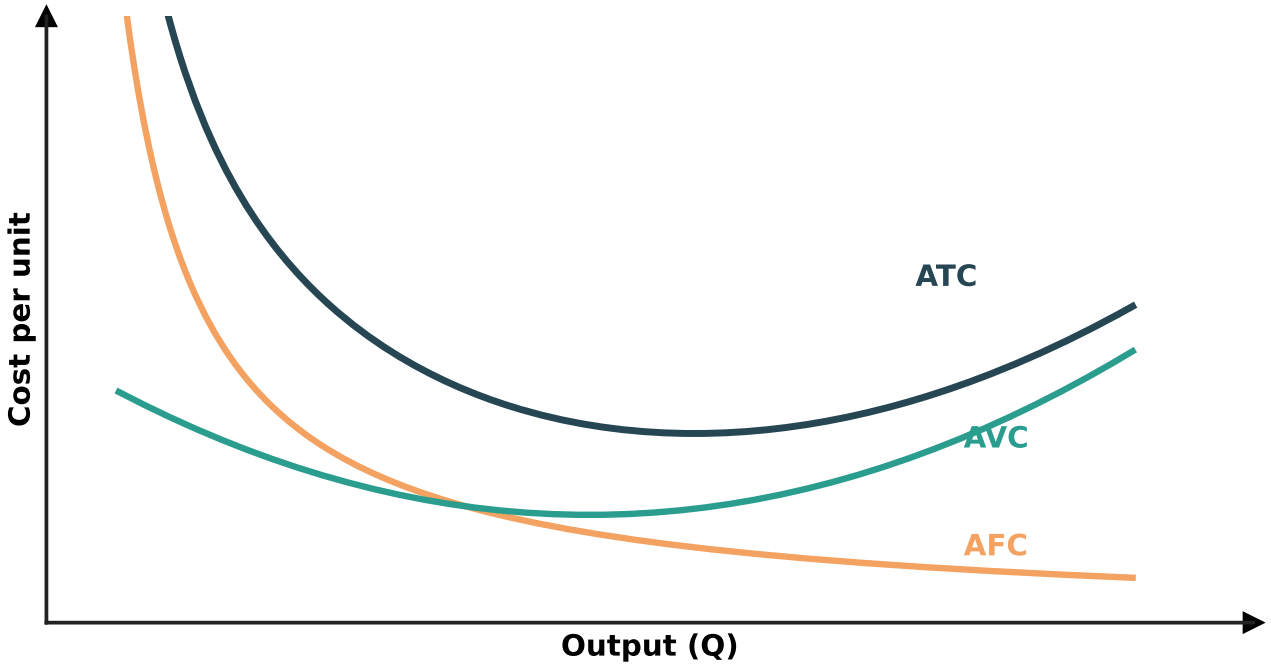
The expansion path joins producer-equilibrium points as output expands (parallel isocost shifts), holding input prices constant. It shows the least-cost input combination for each output level.

18. Total Cost Curves (TC, TFC, TVC)



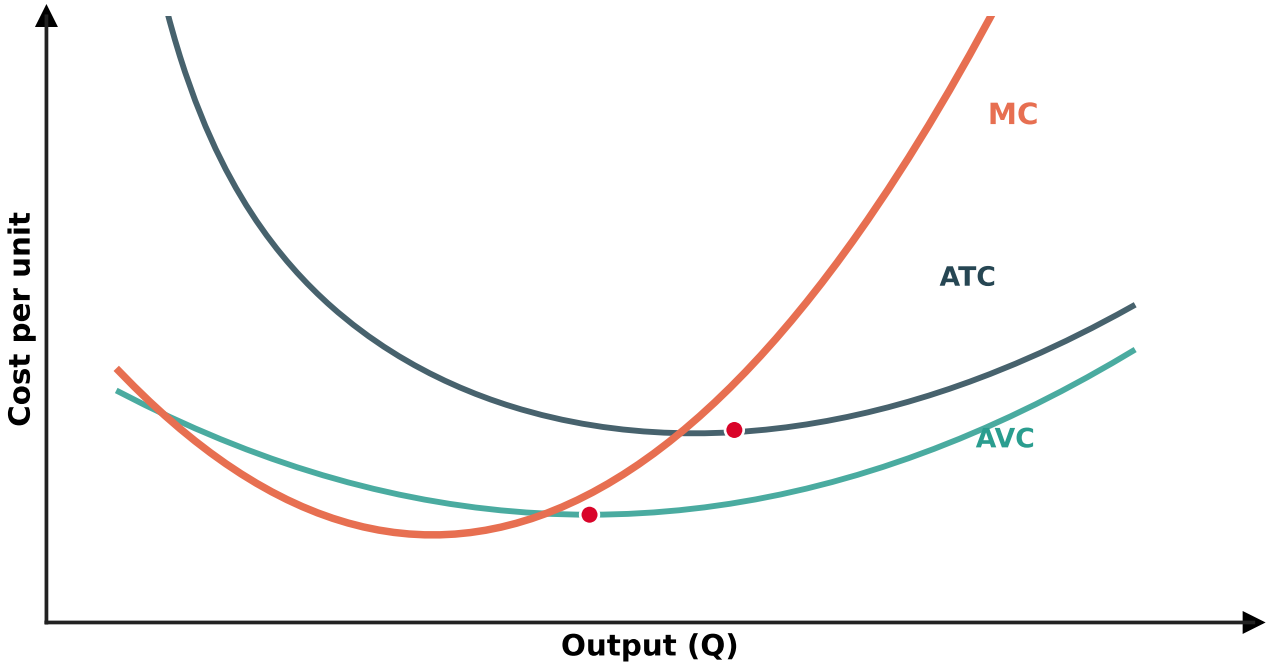
TC = TFC + TVC. TFC is horizontal (independent of output); TVC rises with output (S-shaped). The vertical gap between TC and TVC always equals TFC.

19. Average Cost Curves (AFC, AVC, ATC)



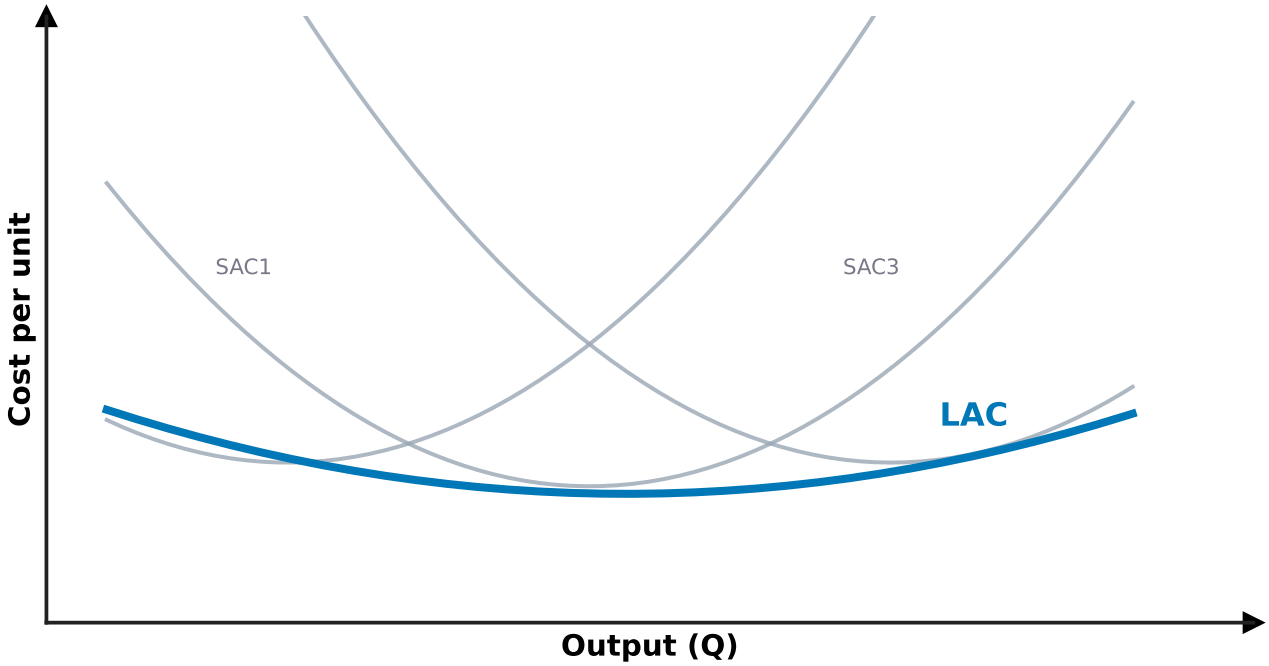
AFC falls continuously (spreading fixed cost). AVC and ATC are U-shaped. The gap between ATC and AVC equals AFC and narrows as output rises.

20. Marginal Cost & U-shaped Costs



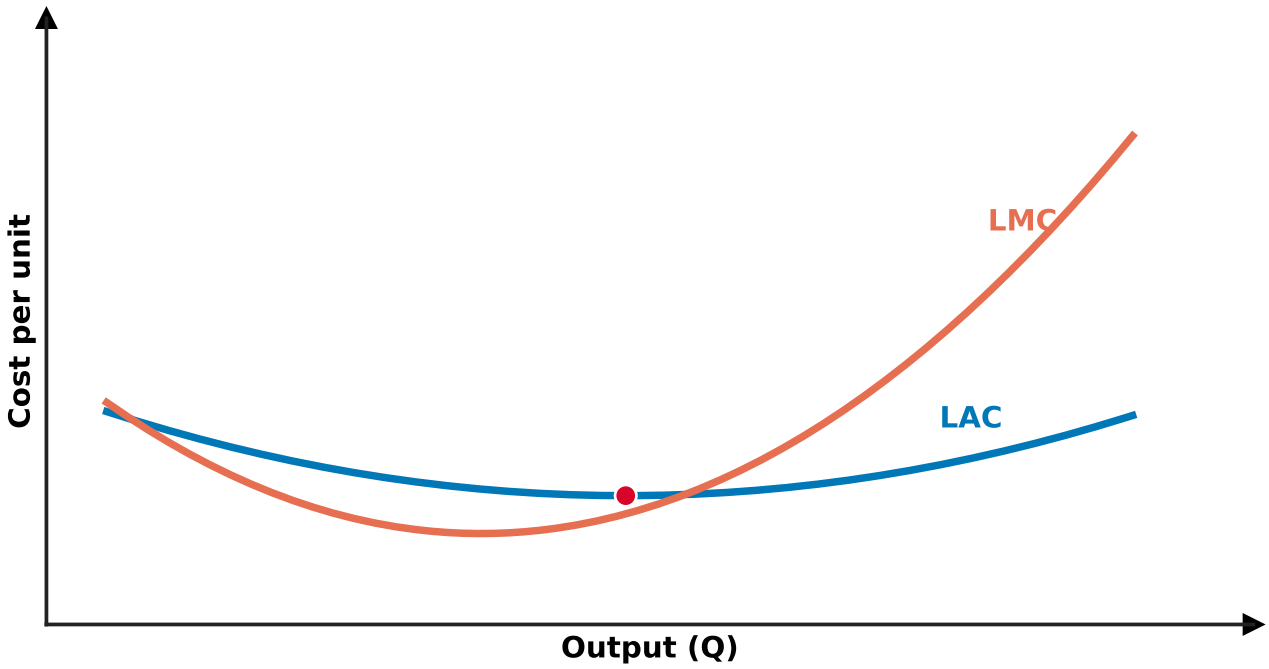
MC is U-shaped and cuts both AVC and ATC at their minimum points (from below). When MC < AC, AC falls; when MC > AC, AC rises — the algebra of averages and margins.

21. Long-Run Average Cost (LAC) — Envelope



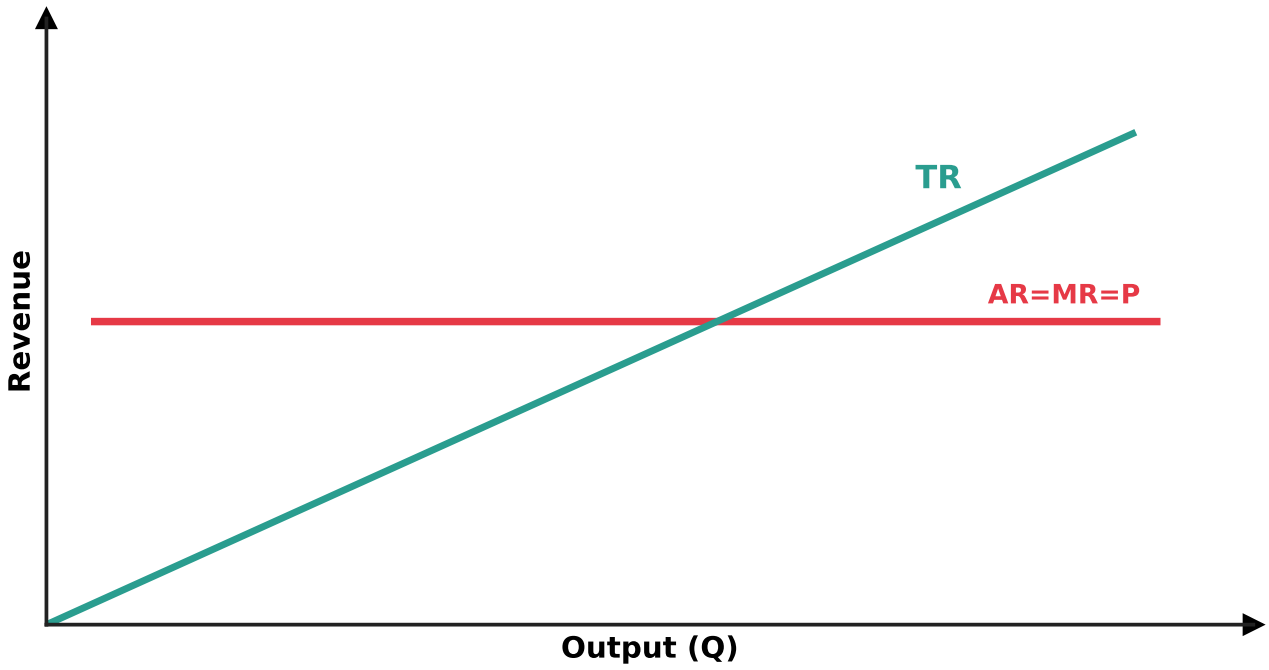
The LAC is the envelope of all short-run ATC curves, showing the least unit cost for each output in the long run. It captures economies of scale (falling part) and diseconomies (rising part).

22. LAC and LMC Relationship



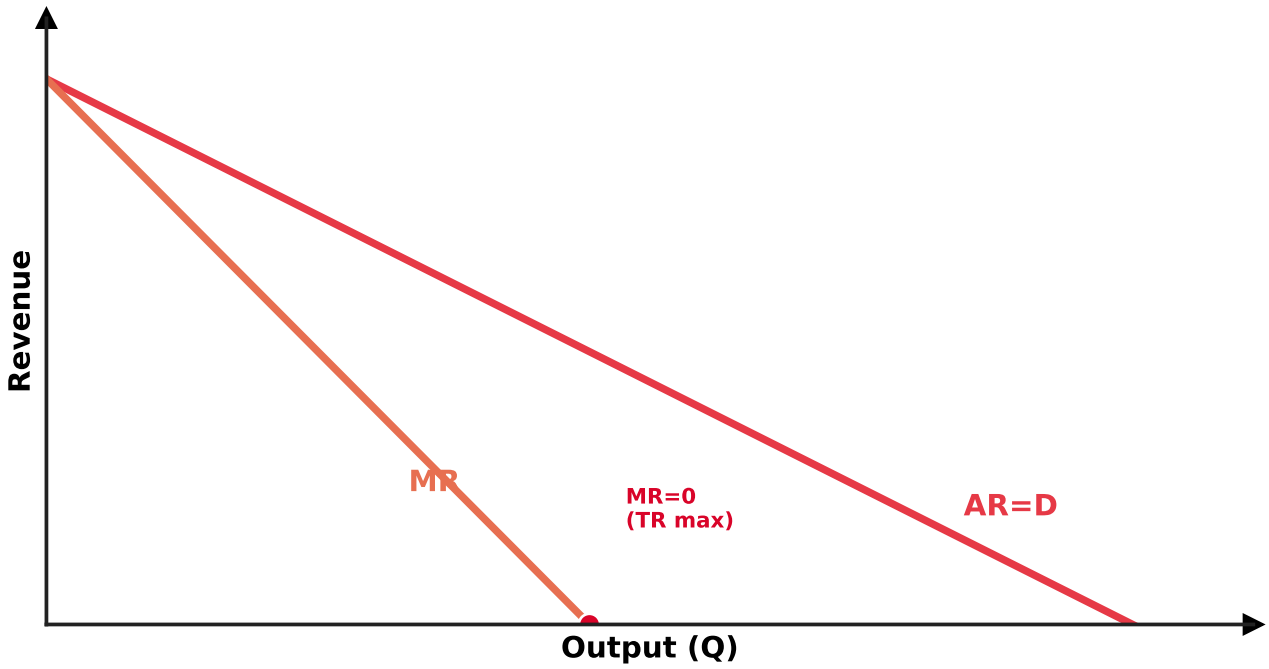
LMC cuts LAC at LAC's minimum point (optimum scale). When LMC < LAC, LAC falls; when LMC > LAC, LAC rises. LMC is flatter than each short-run MC.

23. Revenue under Perfect Competition



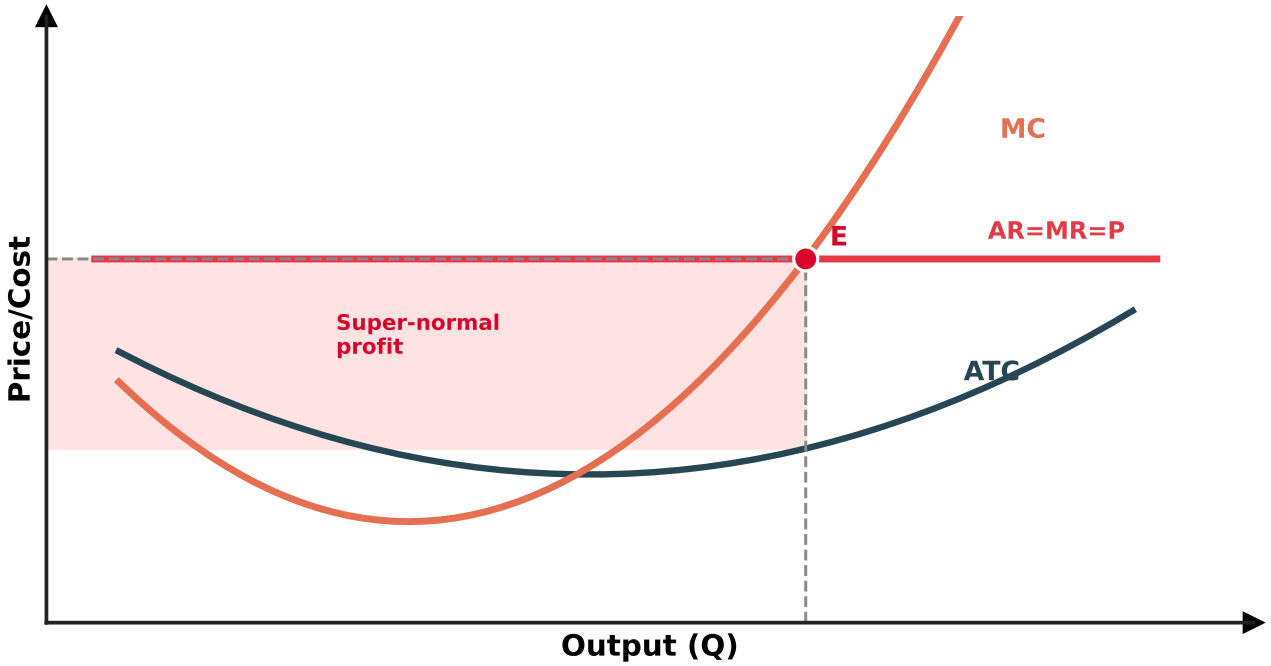
Under perfect competition price is constant, so AR = MR = price (a horizontal line) and TR is a straight upward ray from the origin. The firm is a price-taker.

24. Revenue under Imperfect Competition



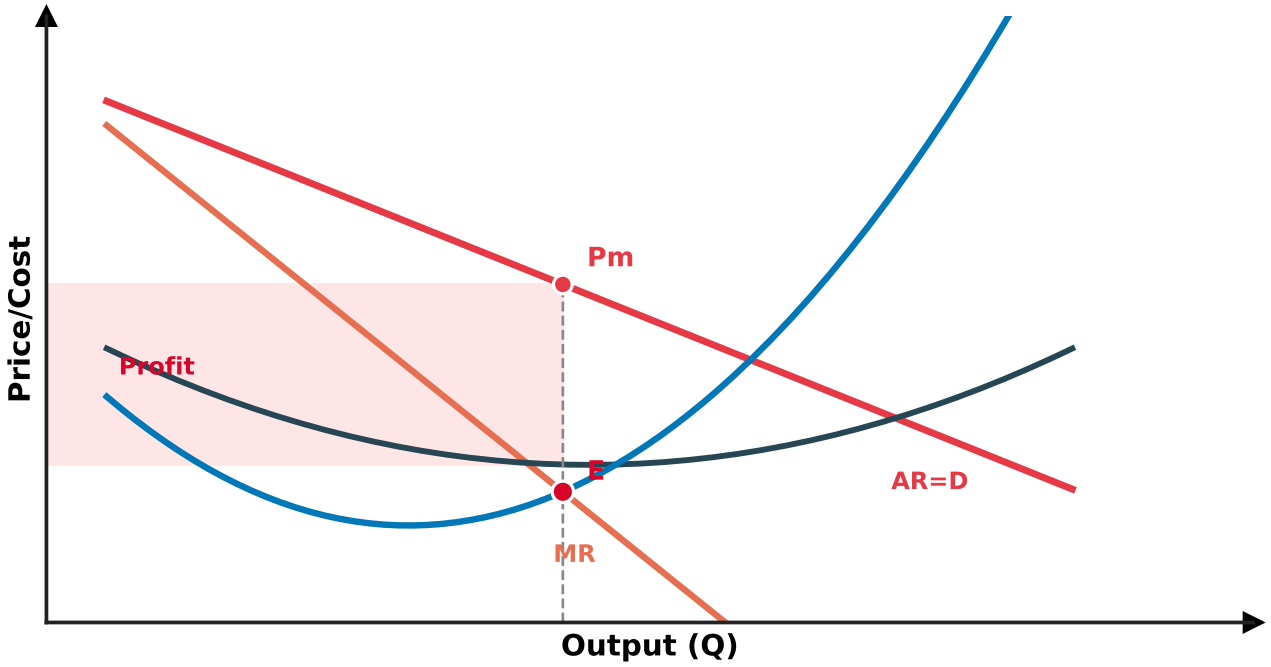
When the demand curve slopes downward (monopoly/monopolistic competition), AR is the demand curve and MR lies below it, falling twice as steeply. TR is maximum where MR = 0.

25. Perfect Competition Equilibrium



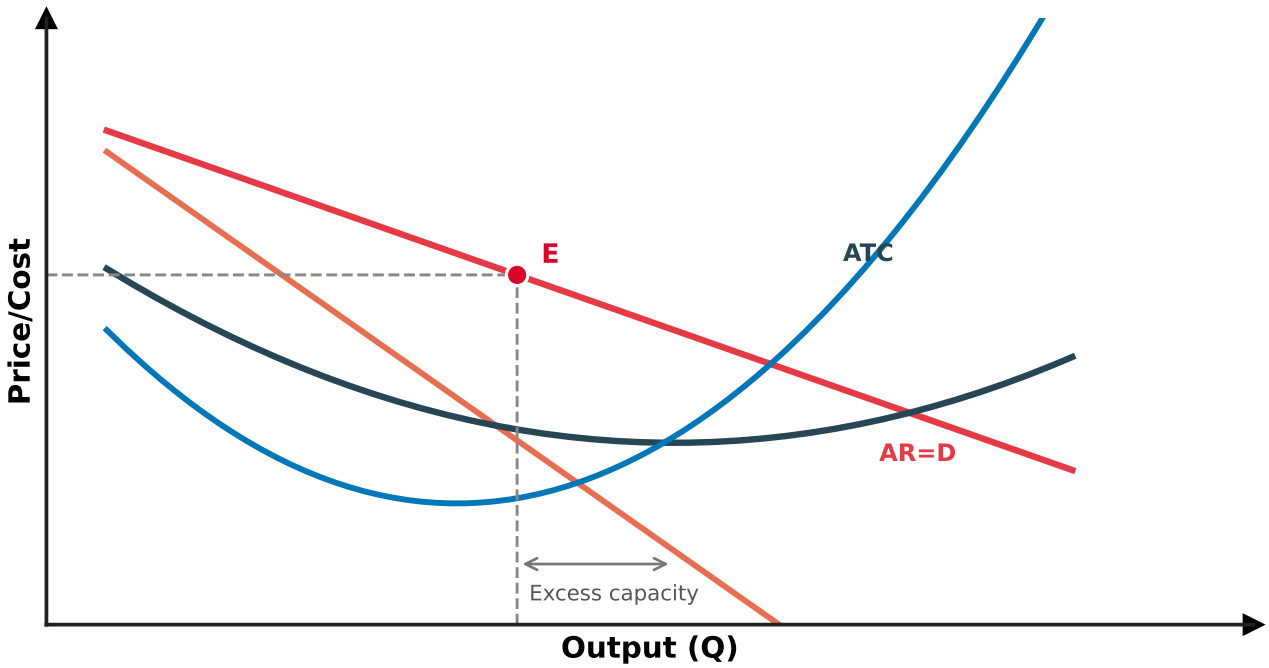
The firm produces where MC = MR = AR = P (point E). With P = ATC it earns normal profit; here P > ATC shows super-normal profit (shaded). Equilibrium requires MC rising and cutting MR from below.

26. Monopoly Equilibrium



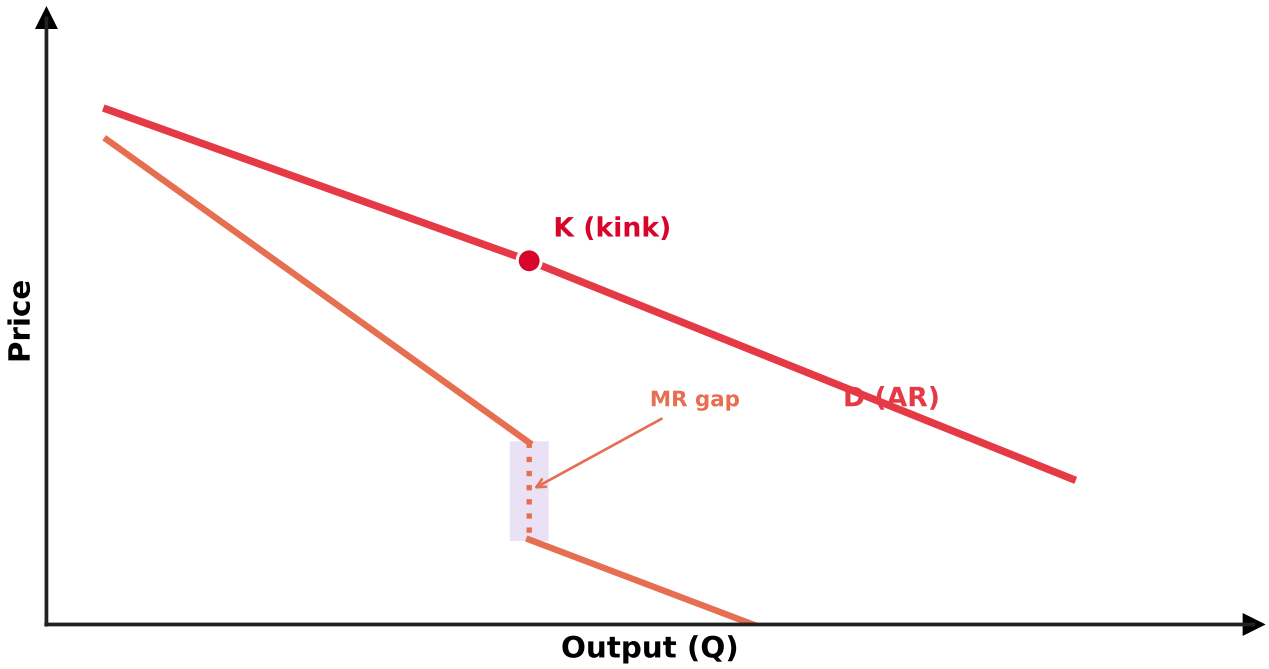
The monopolist equates MC = MR (point E), sets price from the AR/demand curve at Pm (above MR), and earns super-normal profit (shaded). Output is lower and price higher than under perfect competition.

27. Monopolistic Competition (Long Run)



In long-run equilibrium, free entry erodes profit: the demand (AR) curve is tangent to ATC at output where MC = MR. The firm earns only normal profit but operates with excess capacity (left of ATC minimum).

28. Kinked Demand Curve (Oligopoly)



Sweezy's model: rivals match price cuts but ignore price rises, creating a demand curve kinked at the prevailing price (K). The MR curve has a discontinuous gap, so price stays rigid despite MC shifts within the gap.

29. Production Possibility Frontier (PPF)



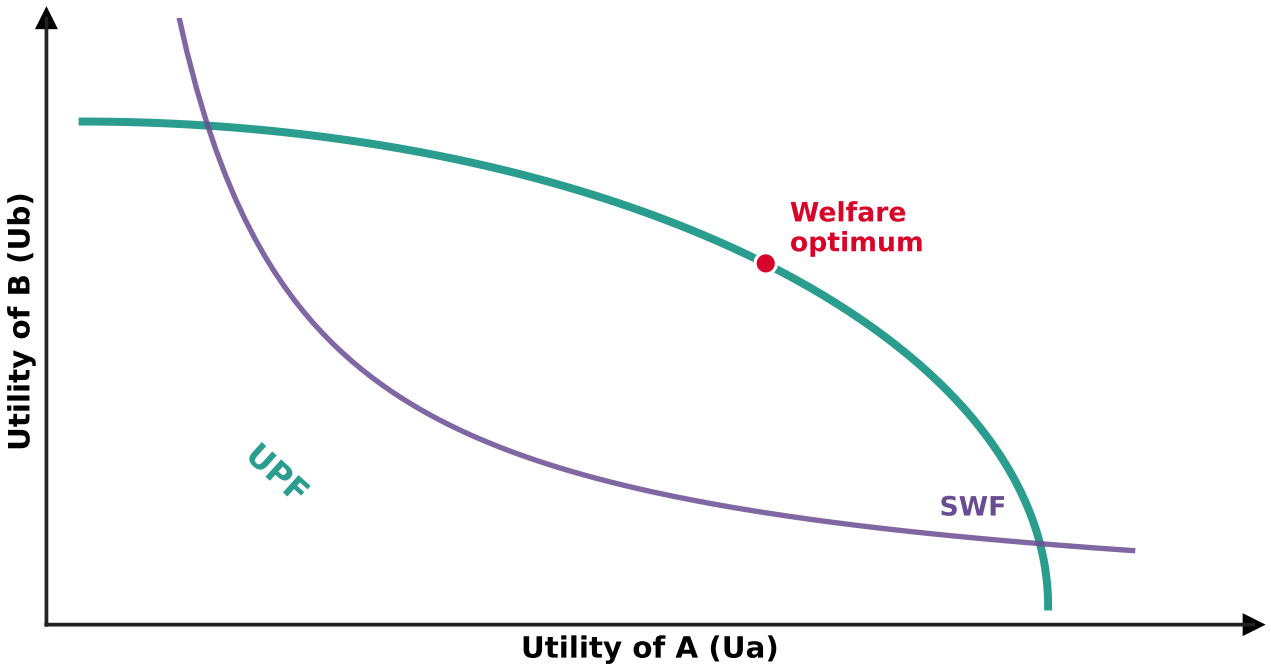
The PPF shows maximum output combinations of two goods with given resources and technology. It is concave (increasing opportunity cost); points inside are inefficient, points outside unattainable.

30. Contract Curve (Edgeworth Box)



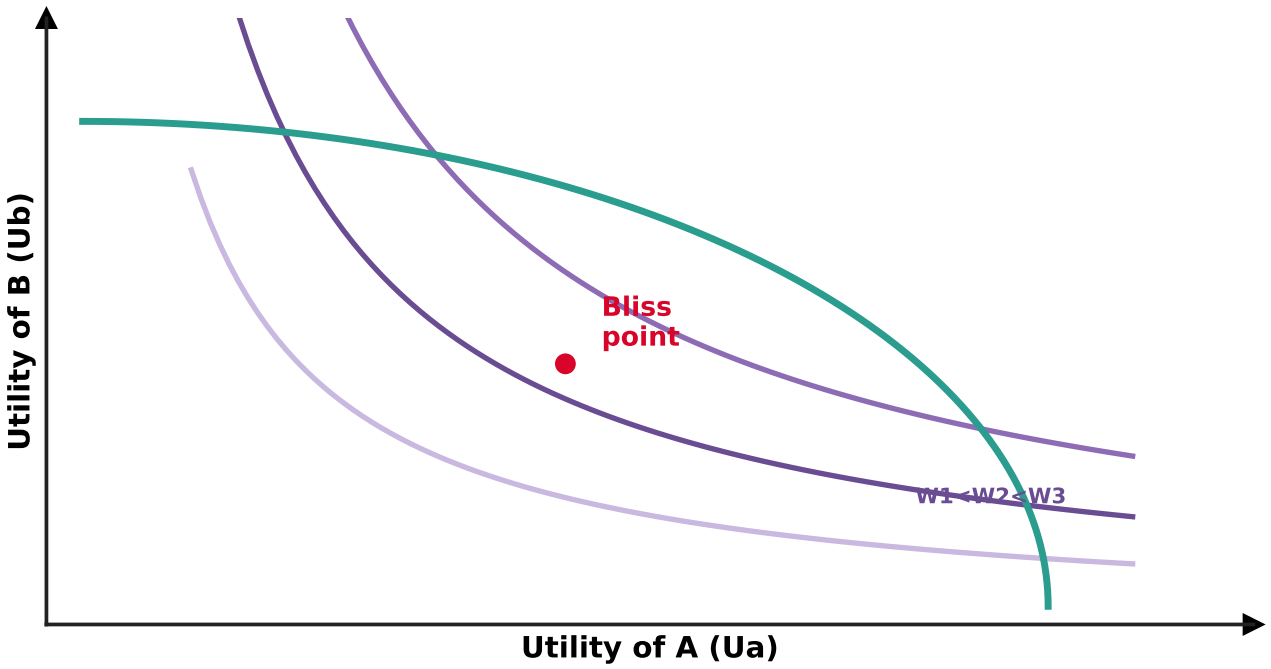
Inside an Edgeworth box the contract curve joins all Pareto-efficient allocations where the two consumers' indifference curves are tangent. Trade moves the economy onto this curve.

31. Utility Possibility Frontier (UPF)



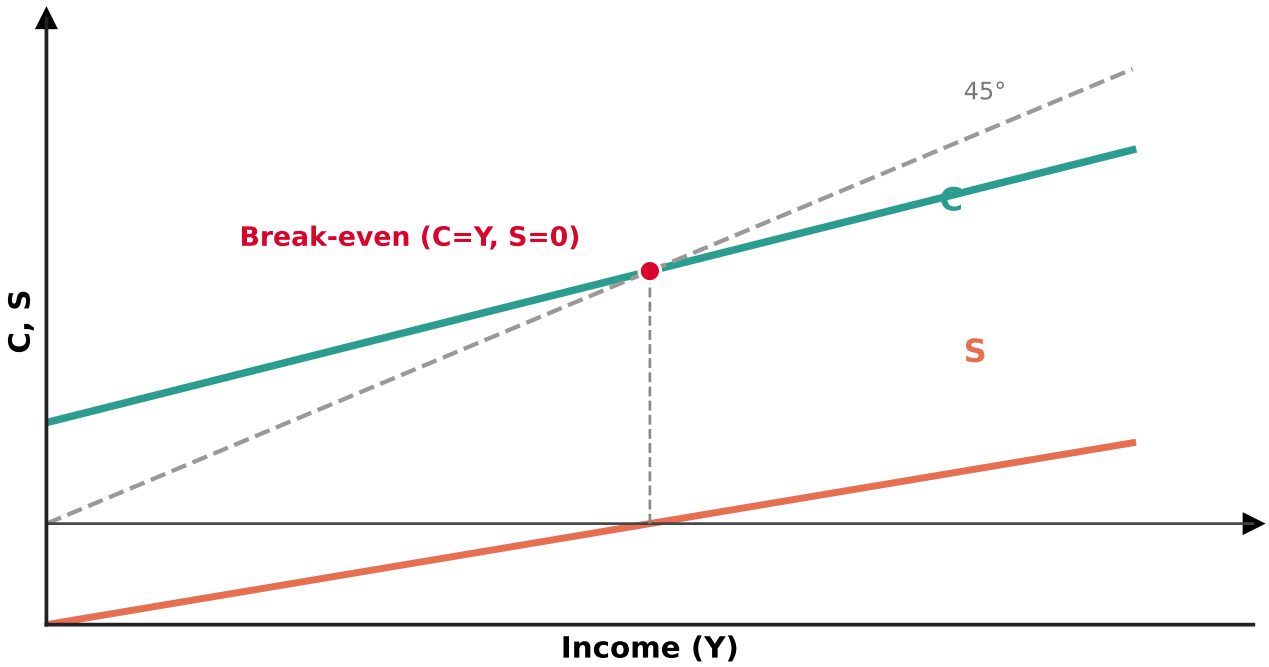
The UPF shows maximum utility attainable by person B for each level given to person A— derived from the contract curve. It is downward-sloping; the welfare optimum is where a social welfare curve is tangent to the UPF.

32. Social Welfare Function (Bergson-Samuelson)



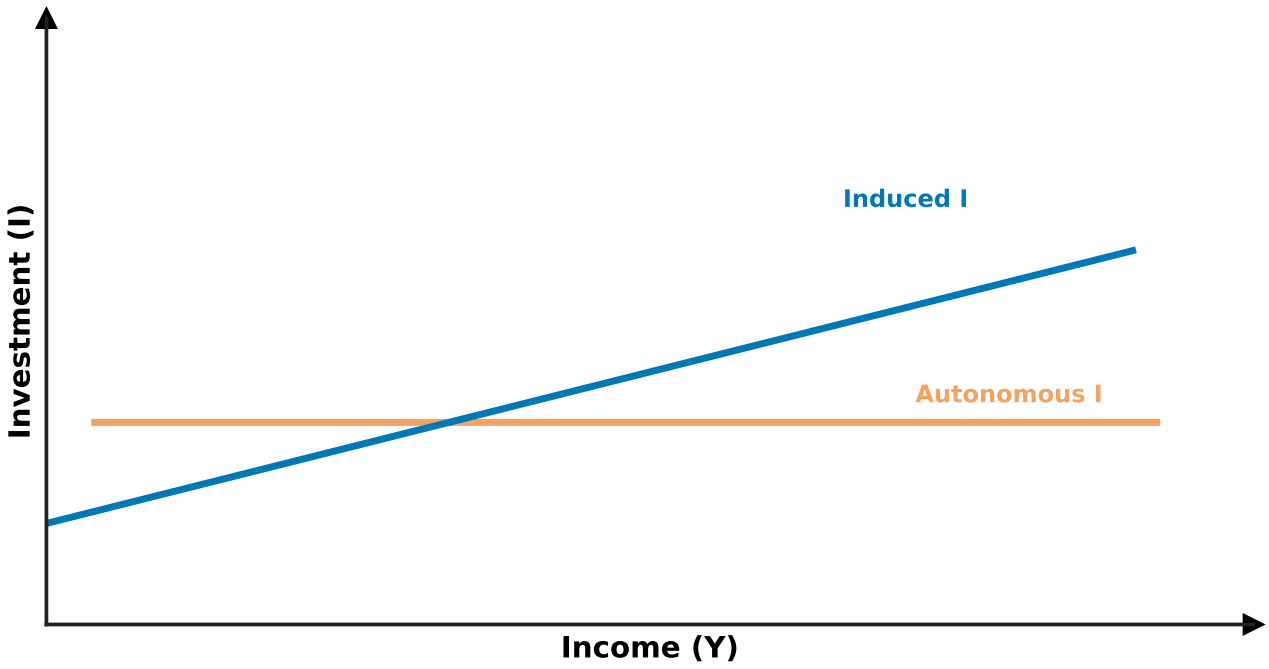
Social indifference (welfare) curves rank distributions of utility between individuals. The point of tangency with the UPF gives the 'bliss point' — the constrained social welfare maximum.

33. Consumption & Saving Functions



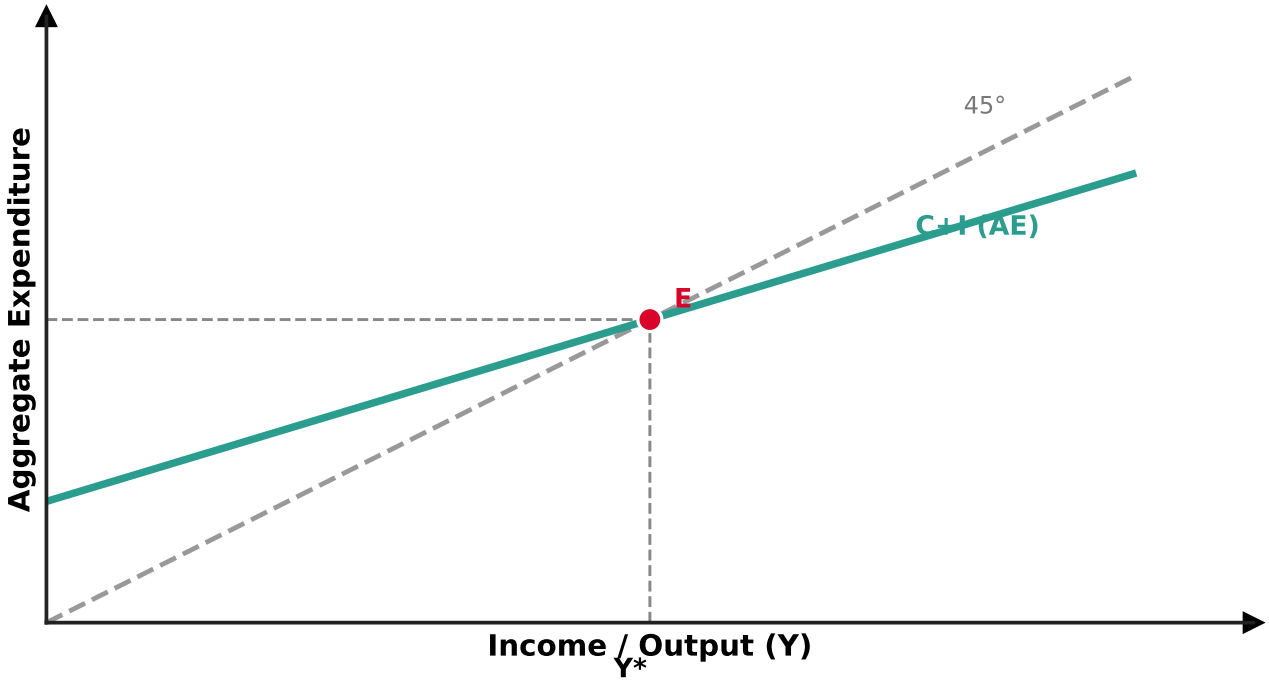
C = a + bY: consumption rises with income but less than proportionately (MPC < 1). Saving S = Y − C is negative below the break-even income (where C = Y) and positive above it. MPC + MPS = 1.

34. Investment Function



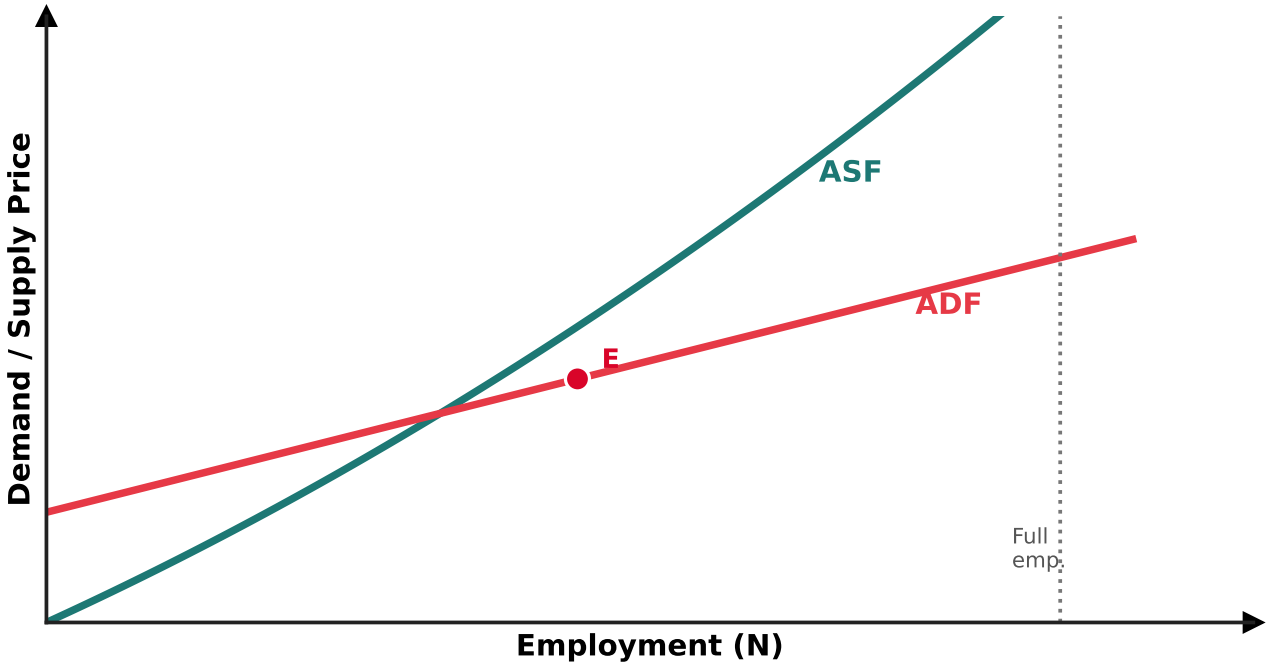
Autonomous investment is independent of income (horizontal line). Induced investment rises with income/output. Investment is also inversely related to the rate of interest (MEC schedule).

35. Keynesian Cross (Income Determination)



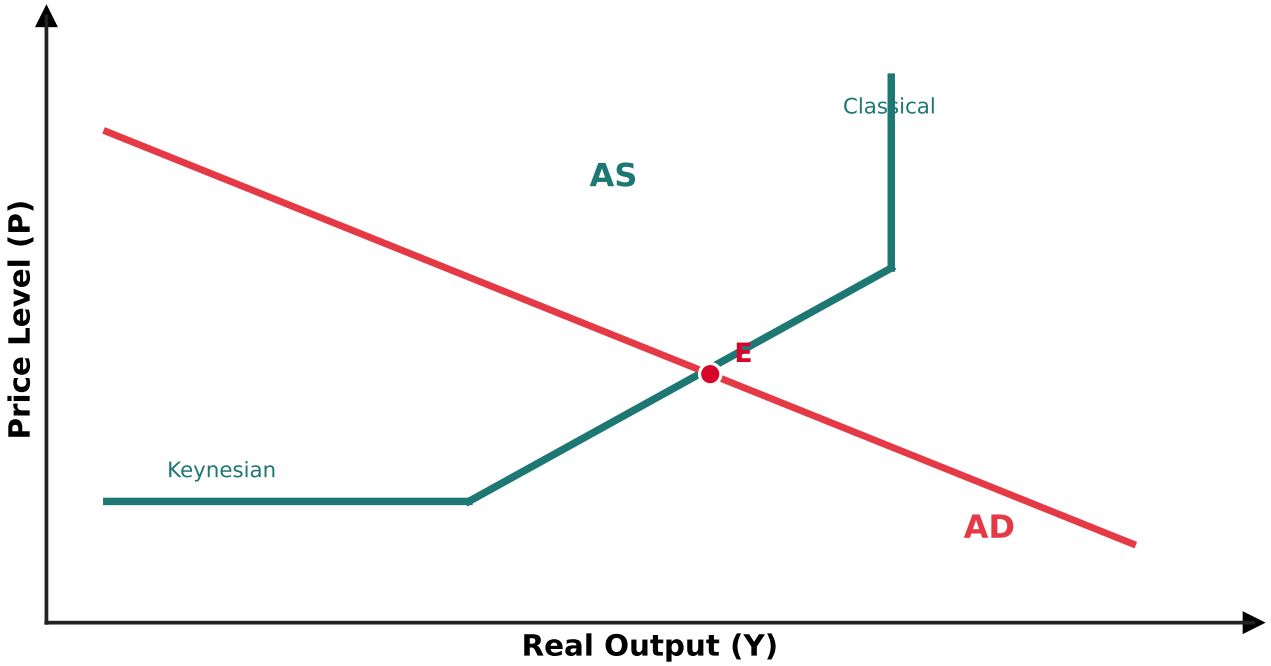
Equilibrium income is where aggregate expenditure (C + I) crosses the 45° line (E), i.e. AE = Y. Excess demand above and deficient demand below this point drive output to equilibrium Y*.

36. Effective Demand (AD–AS)



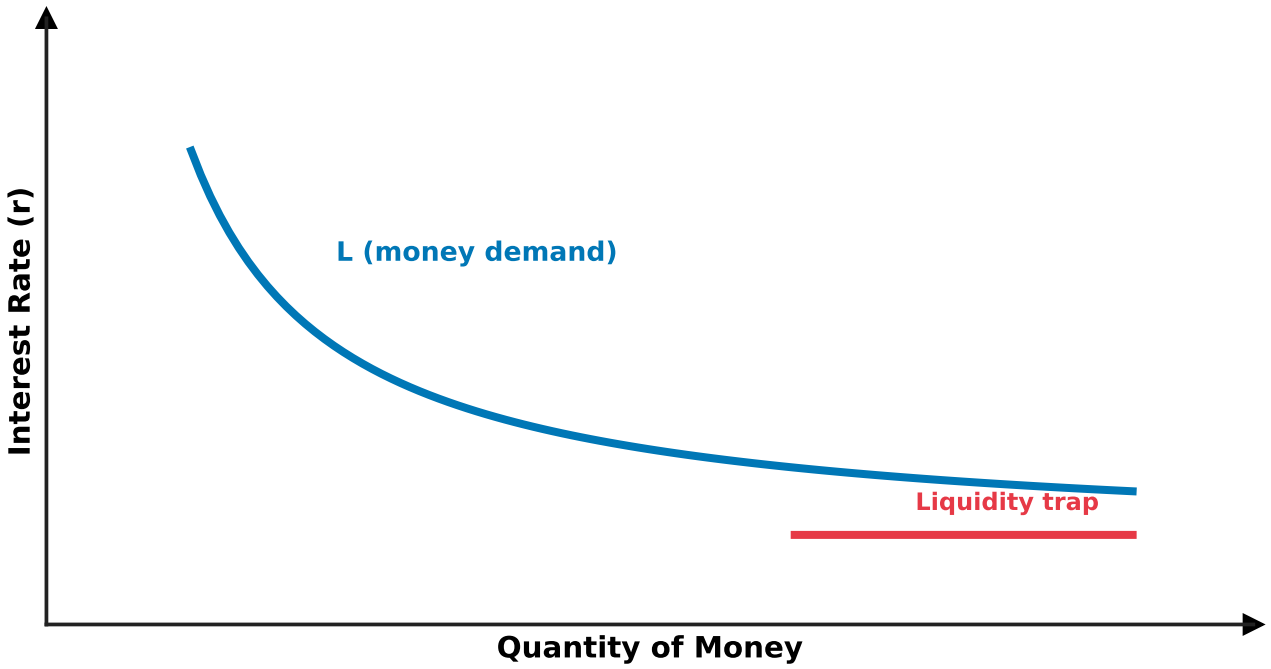
Effective demand is the level of output where the aggregate demand price equals the aggregate supply price (E). Keynes argued equilibrium can occur below full employment, leaving an unemployment gap.

37. Aggregate Demand & Supply (AD–AS Model)



The AS curve has a flat Keynesian range, an upward intermediate range, and a vertical classical (full-employment) range. Equilibrium price and output are set where AD intersects AS.

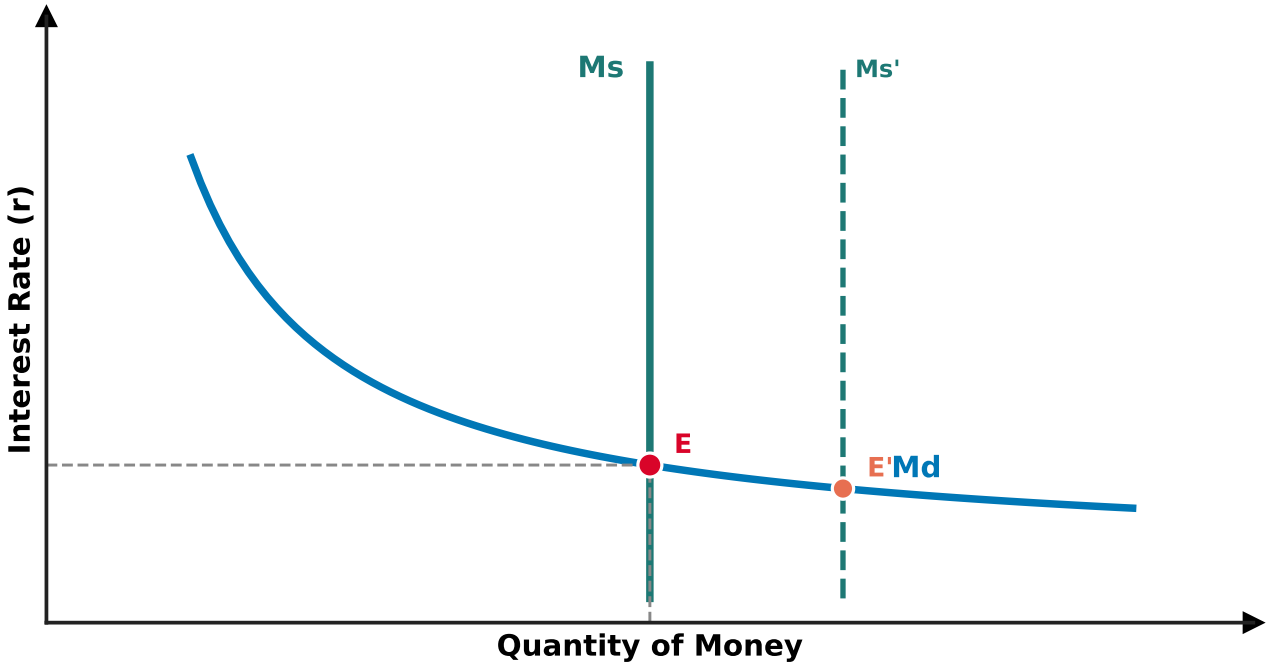
38. Liquidity Preference (Money Demand)



Demand for money (L) falls as the interest rate rises (transactions + speculative motives). At very low rates the curve becomes horizontal — the liquidity trap, where monetary policy loses traction.

10. Money & Banking

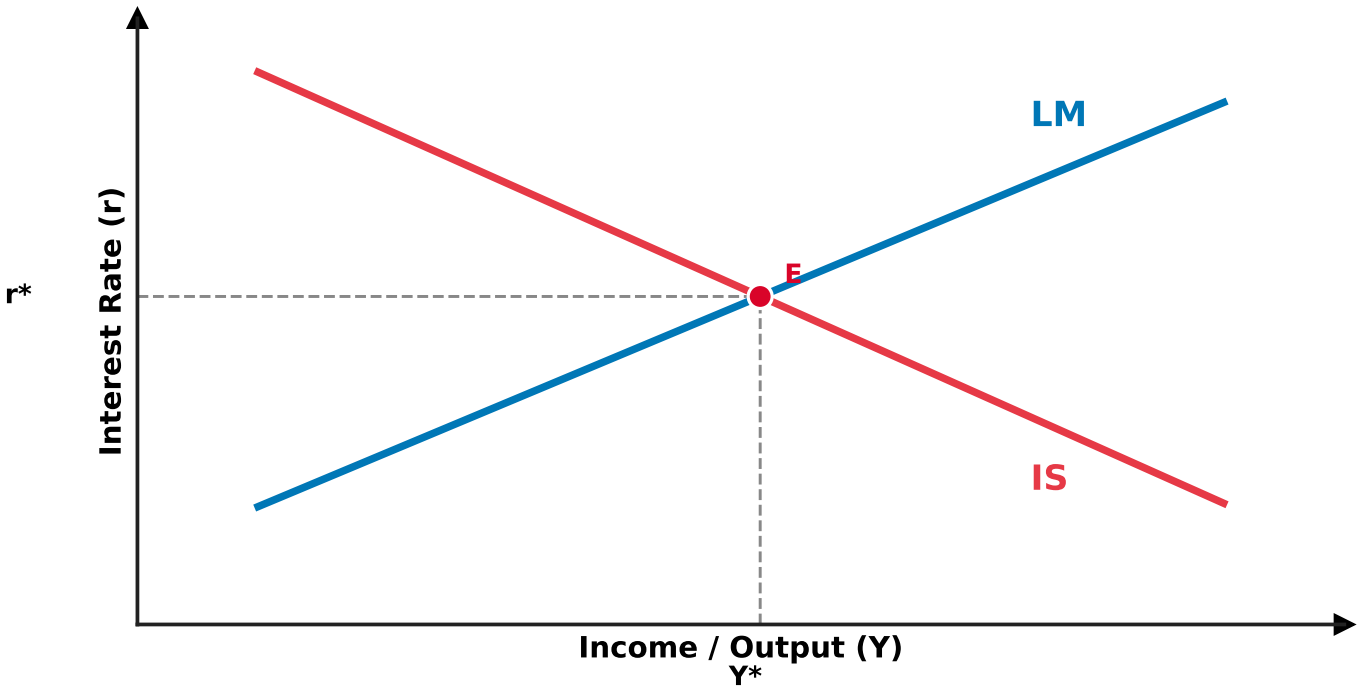
39. Money Market Equilibrium



Money supply (Ms) is vertical (set by the central bank). Equilibrium interest rate r* is where Ms meets money demand (Md). A rise in Ms shifts the line right and lowers r*.

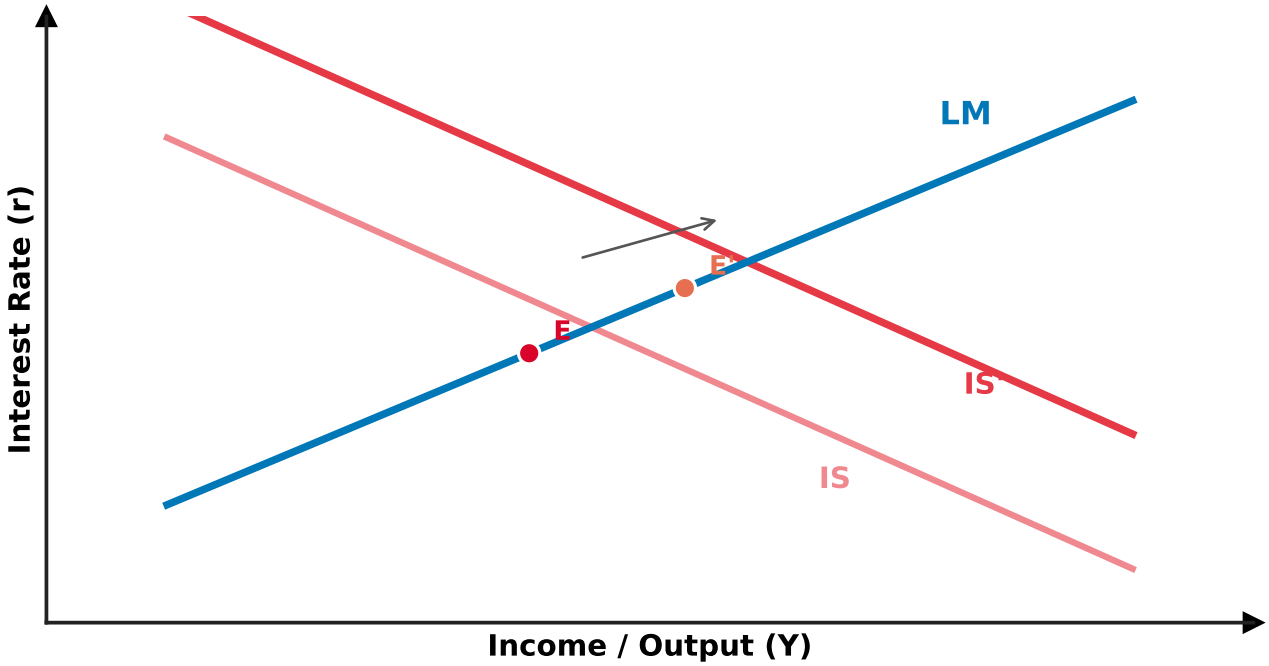
11. IS-LM Model

40. IS-LM Equilibrium



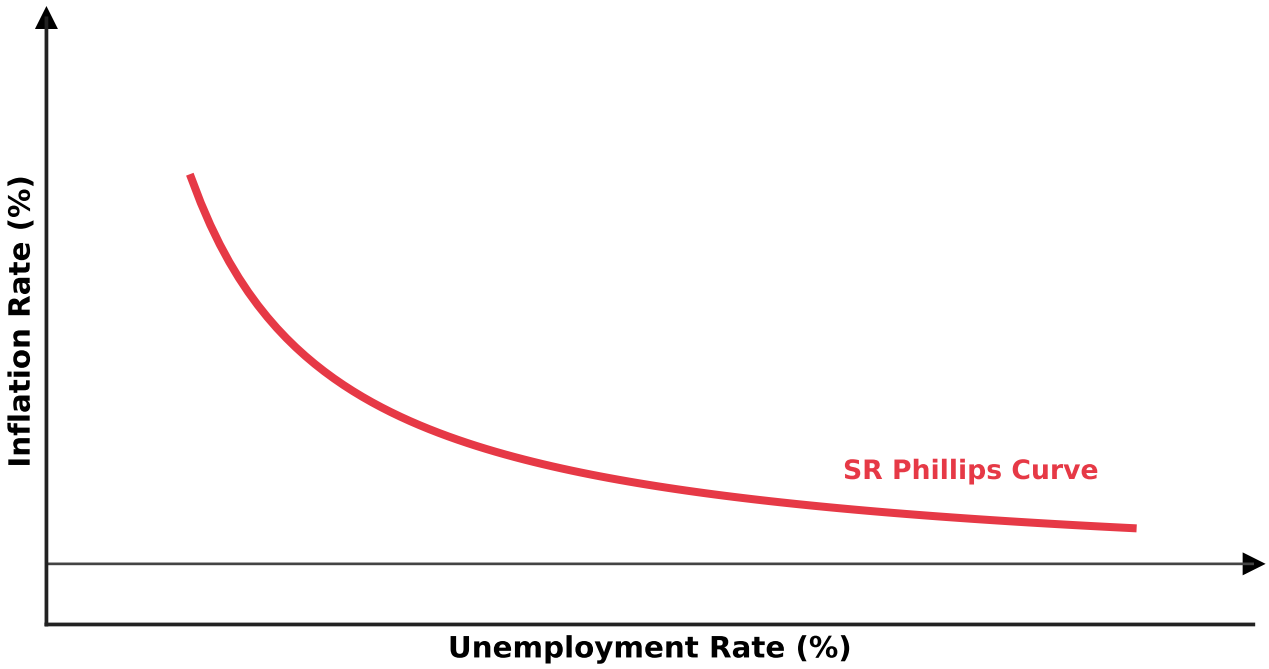
The IS curve (goods market) slopes down; the LM curve (money market) slopes up. Their intersection (E) gives simultaneous equilibrium income Y* and interest rate r*. Fiscal policy shifts IS; monetary policy shifts LM.

41. Fiscal Expansion in IS-LM



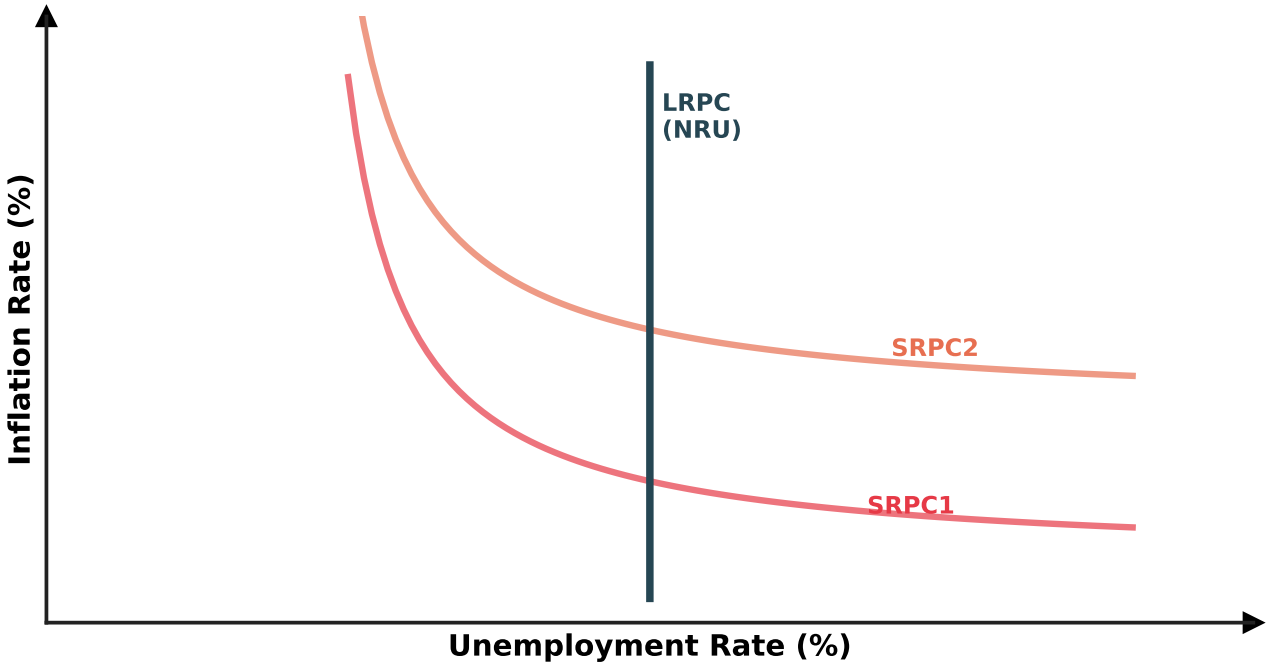
Expansionary fiscal policy shifts IS right (IS→IS'), raising both income and interest rate. The interest-rate rise partly 'crowds out' private investment, so income rises less than the full multiplier amount.

42. Phillips Curve (Short Run)



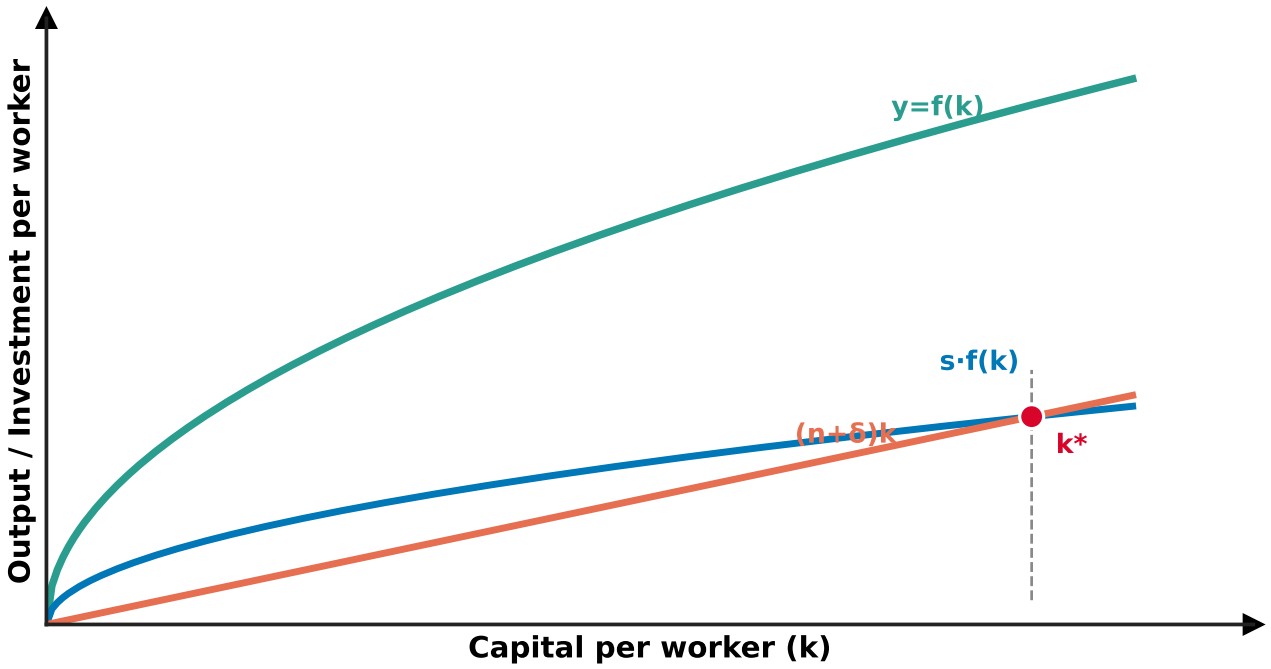
The original Phillips curve shows an inverse trade-off between unemployment and inflation: lower unemployment is associated with higher inflation. It suggests policymakers can 'buy' lower unemployment with more inflation.

43. Expectations-Augmented / Long-Run Phillips Curve



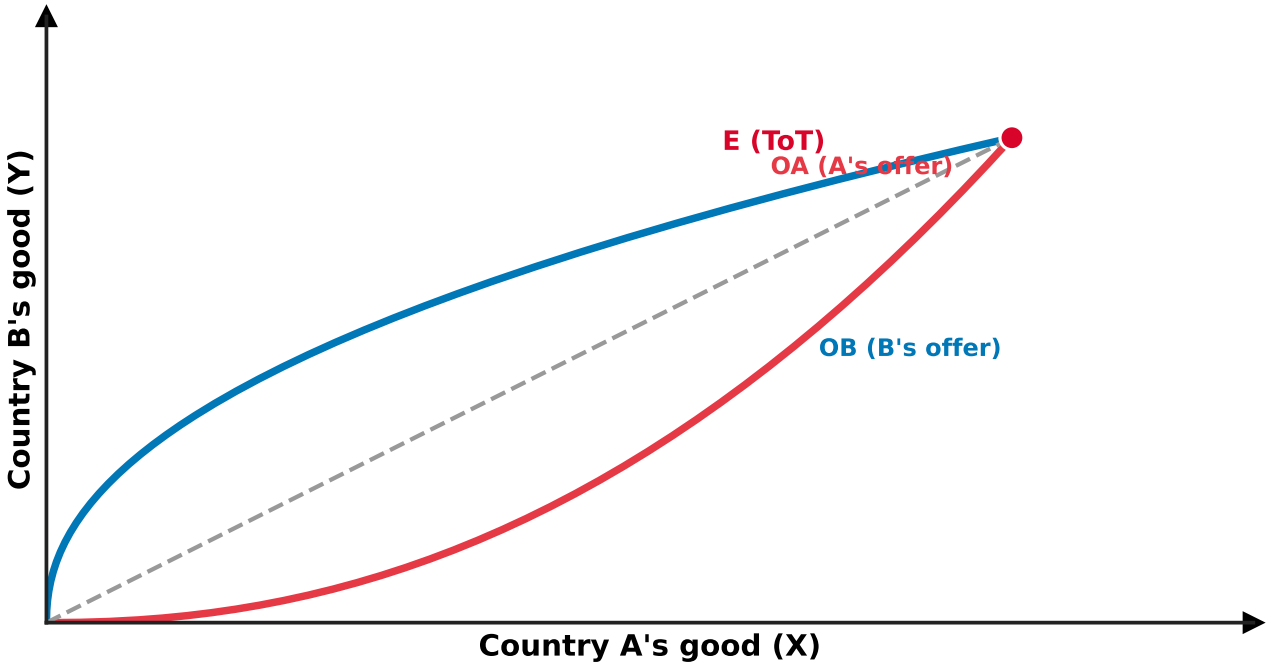
Friedman–Phelps: as inflation expectations adjust, short-run curves shift up. In the long run the curve is vertical at the natural rate of unemployment (NRU) — no permanent inflation–unemployment trade-off.

44. Solow Growth Model



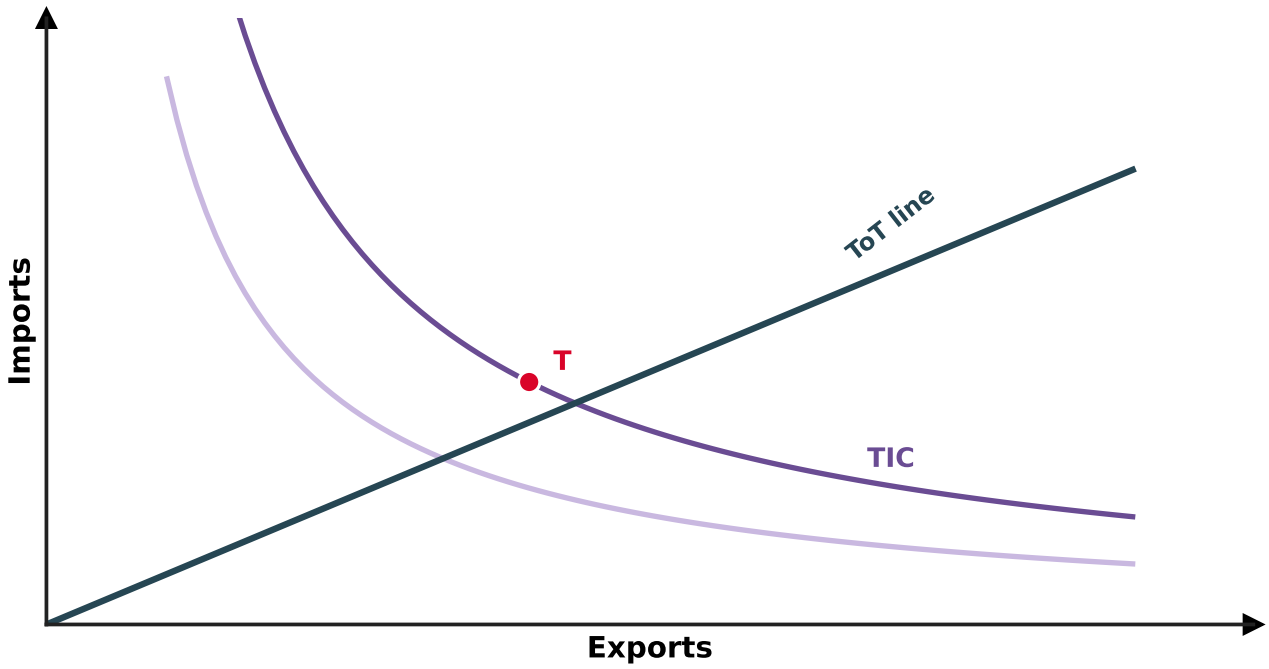
Per-worker output y = f(k); investment s·f(k) and break-even (n+δ)k determine the steady state k* where they intersect. At k* capital per worker is constant; long-run growth depends on technological progress.

45. Offer Curve (Reciprocal Demand)



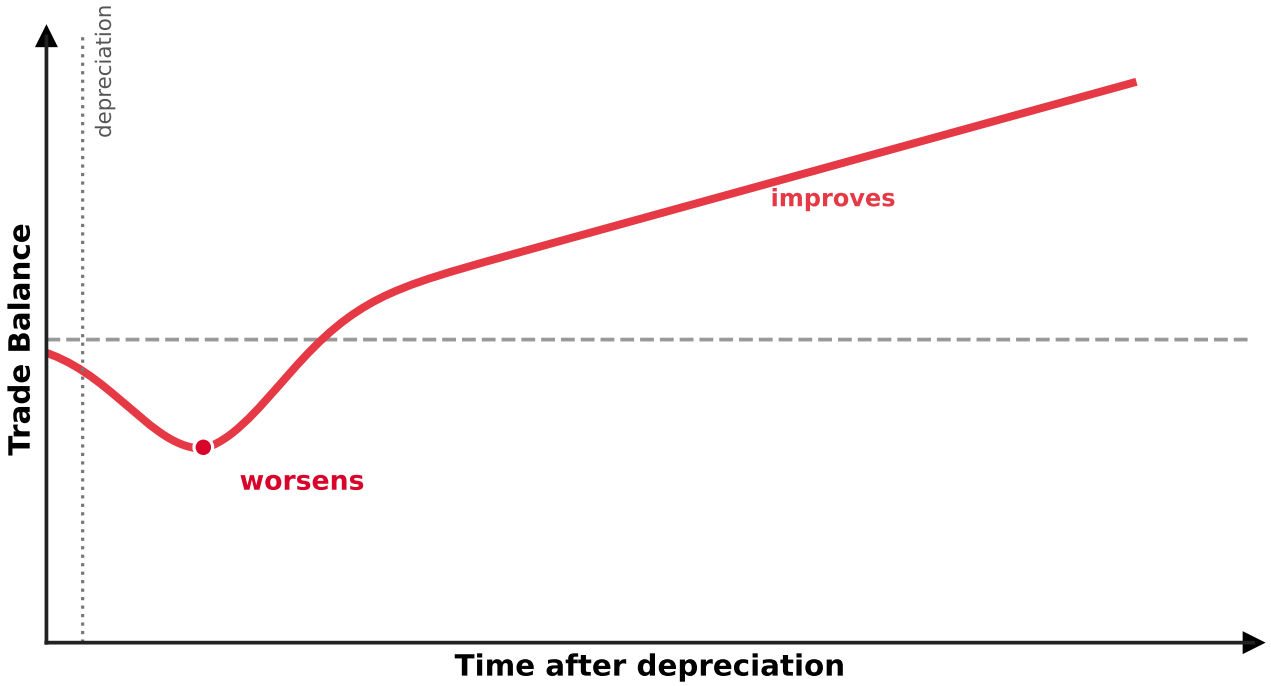
An offer curve shows quantities of exports a country offers for given imports. The intersection of two countries' offer curves determines the equilibrium terms of trade (the slope of the line from the origin).

46. Terms of Trade & Trade Indifference



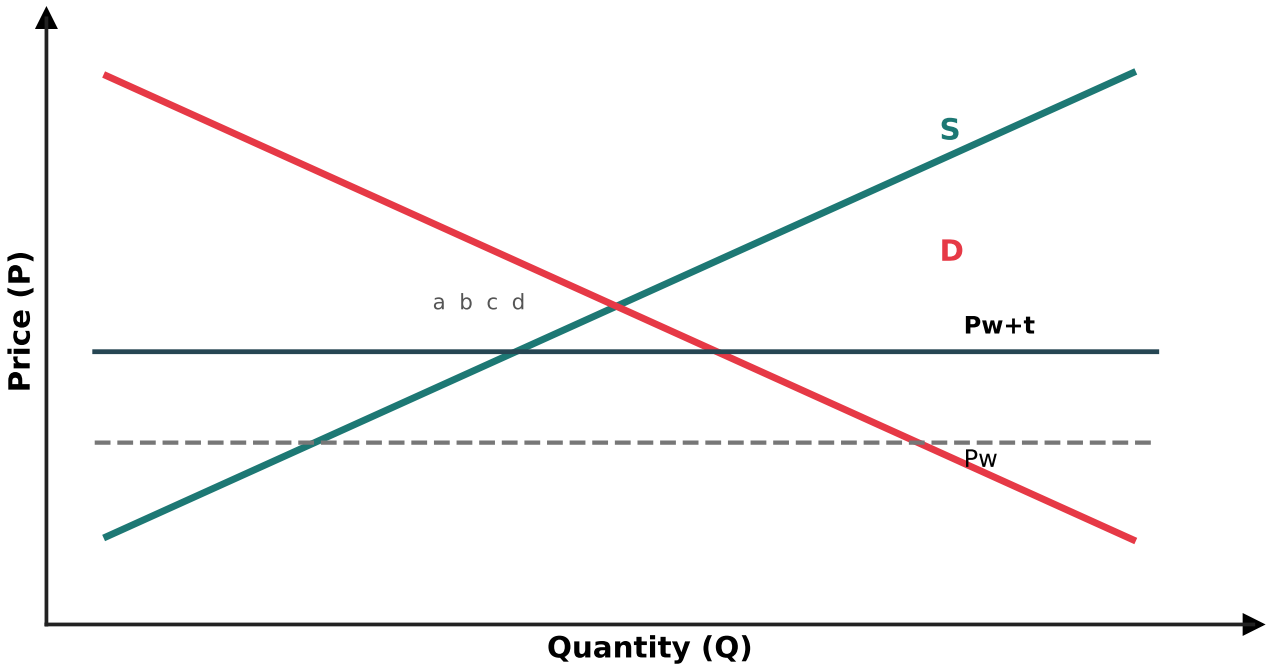
Trade indifference curves represent combinations of trade giving equal welfare. The terms-of-trade line tangent to the highest attainable trade indifference curve gives the country's optimal trade point.

47. J-Curve Effect



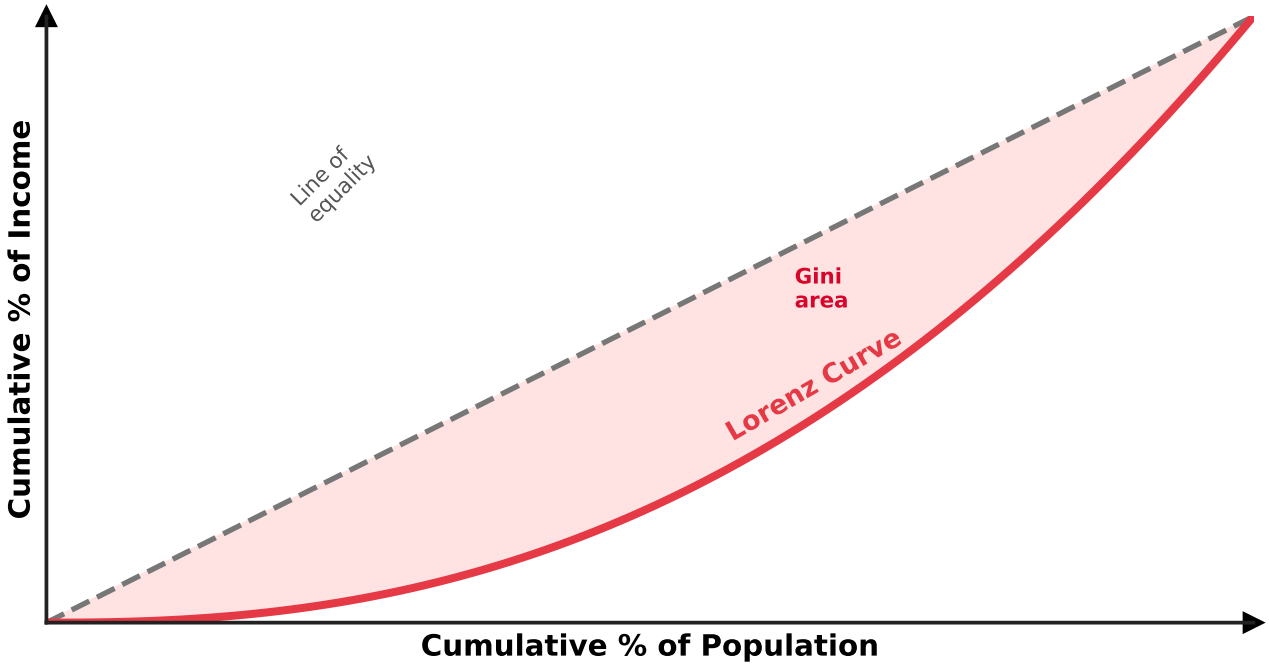
After a currency depreciation the trade balance first worsens (existing contracts, sticky volumes) then improves as export and import quantities adjust — tracing a 'J' shape over time (Marshall–Lerner condition).

48. Tariff: Welfare Effects



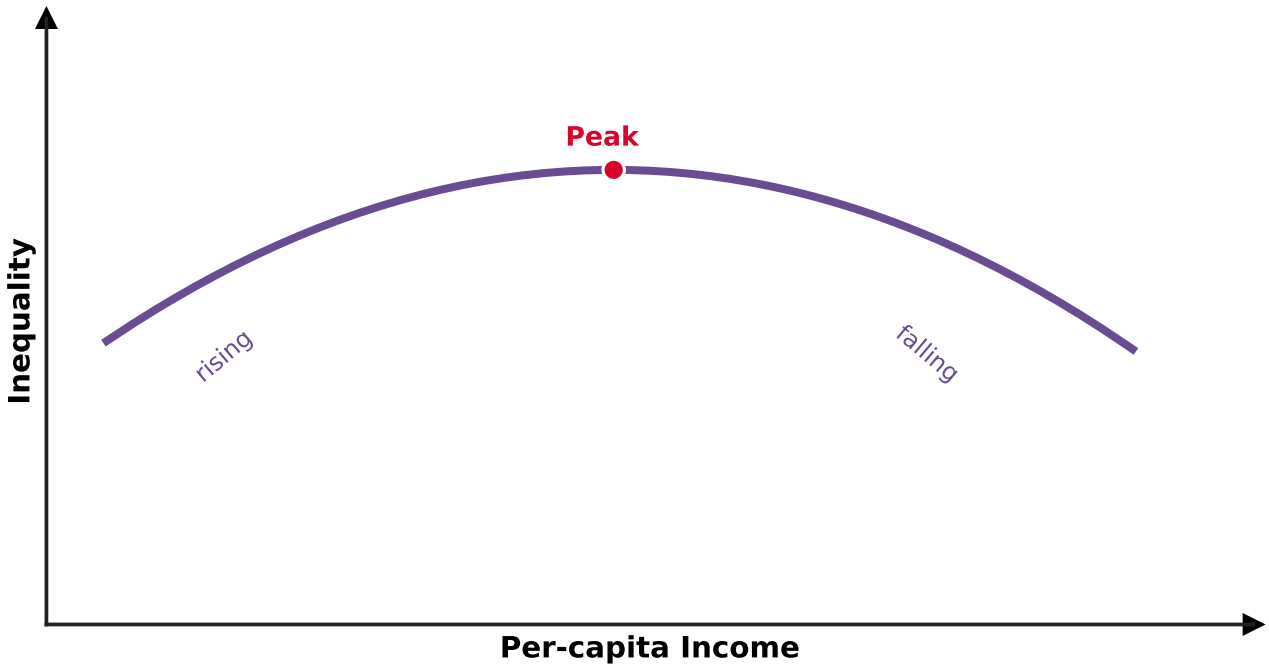
A tariff raises domestic price from Pw to Pw+t, cutting imports. Consumers lose; producers and government (tariff revenue) gain. The two triangles (b + d) are the deadweight efficiency loss to the economy.

49. Lorenz Curve



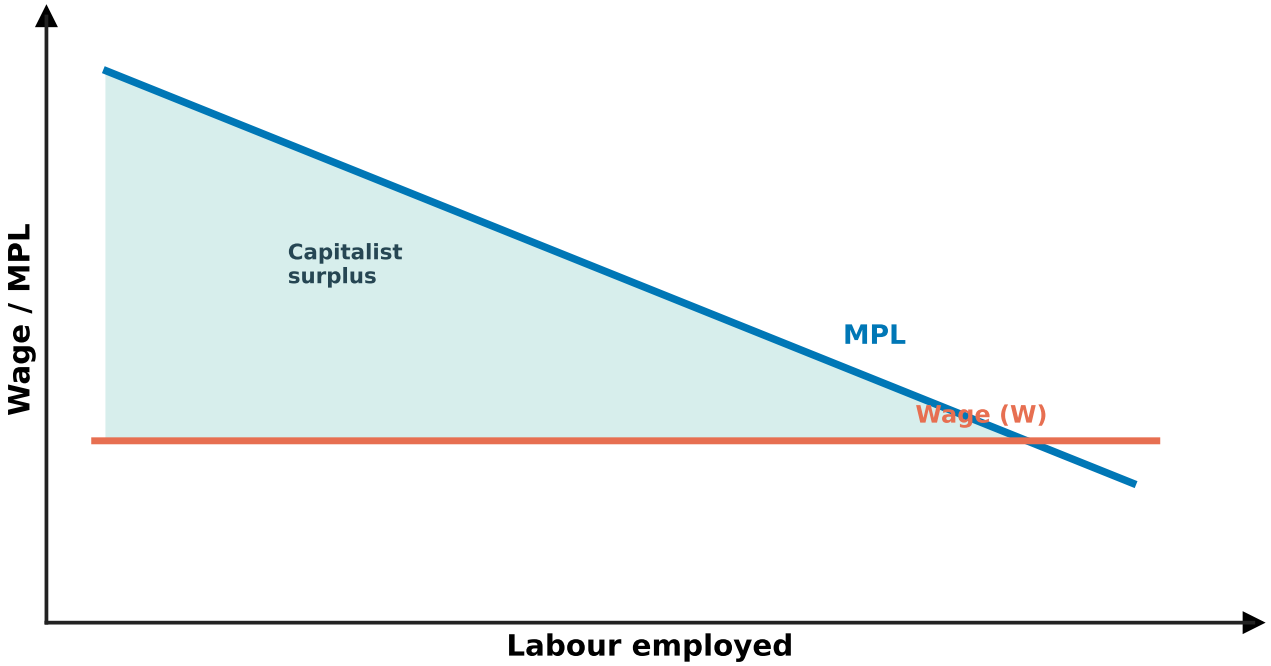
The Lorenz curve plots cumulative share of income against cumulative share of population. The further it bows from the 45° line of perfect equality, the greater the inequality (measured by the Gini coefficient).

50. Kuznets Curve



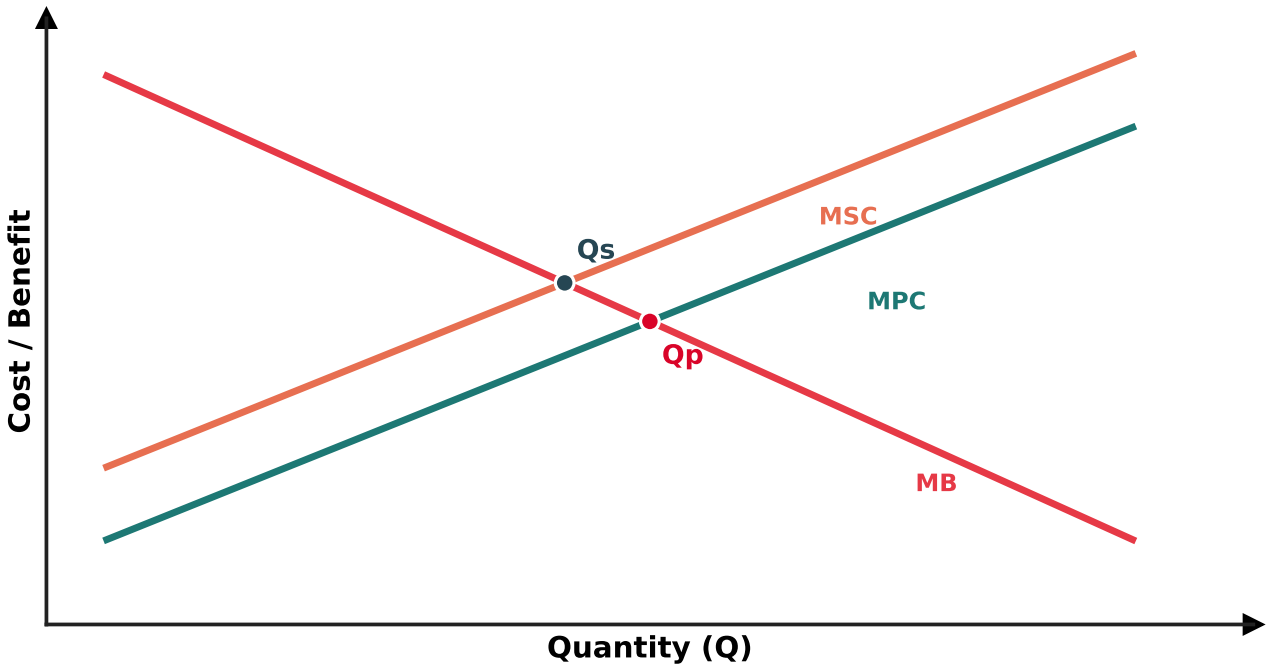
Kuznets' inverted-U hypothesis: income inequality first rises in early development, peaks, then falls as the economy matures. Inequality is plotted against per-capita income.

51. Lewis Model (Surplus Labour)



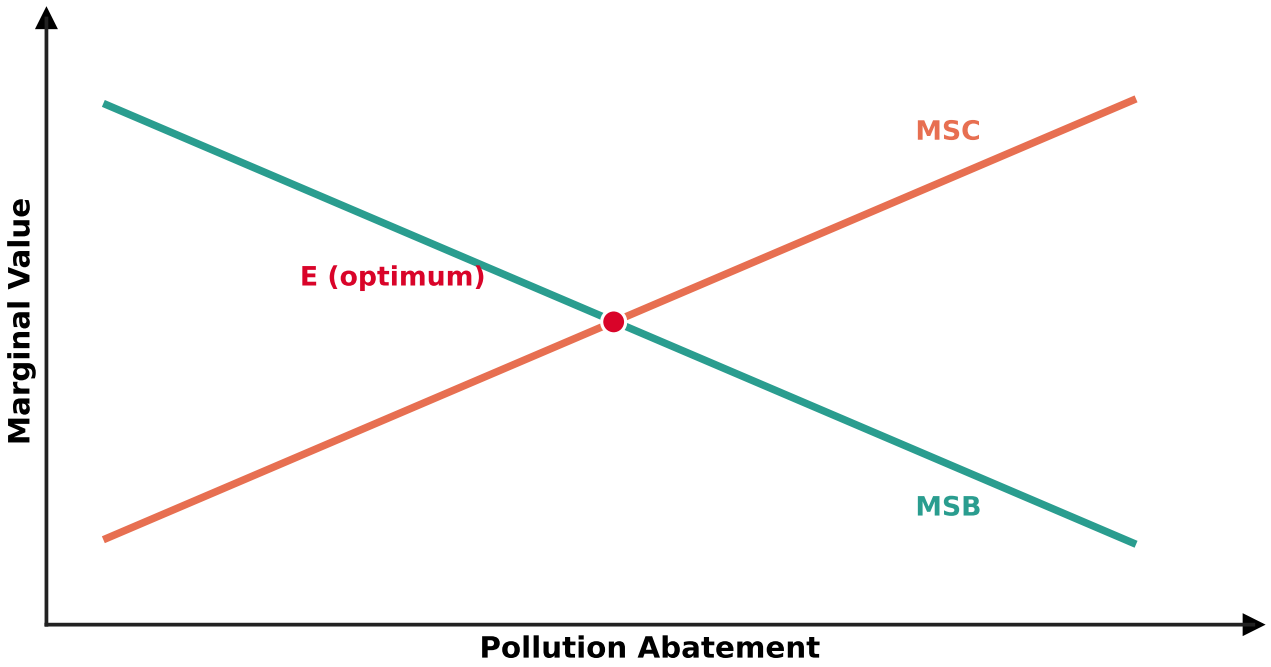
In the dual economy, the modern sector draws labour from agriculture at a constant wage while marginal product exceeds the wage, generating reinvestable surplus (shaded) until the 'Lewis turning point' is reached.

52. Externality: Private vs Social Cost



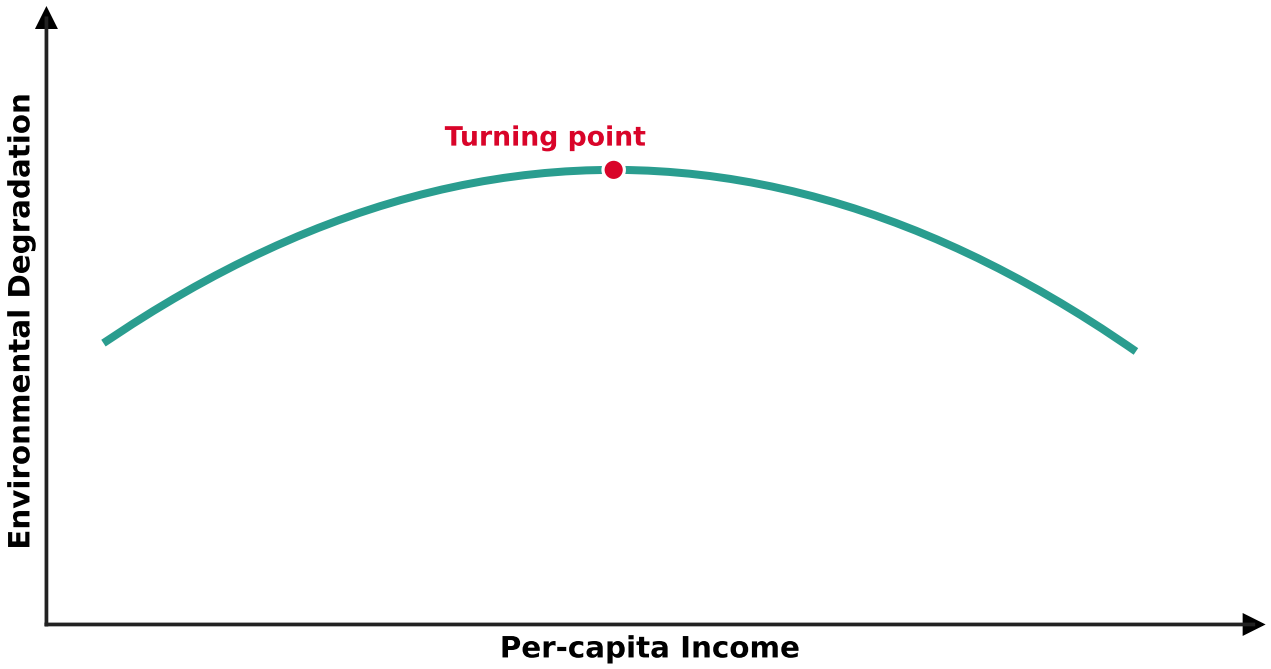
With a negative externality, Marginal Social Cost exceeds Marginal Private Cost. The market produces at Qp (MPC=MB) but the social optimum is Qs (MSC=MB). The shaded triangle is the welfare loss from over-production.

53. Marginal Social Benefit vs Cost (Optimal Pollution)



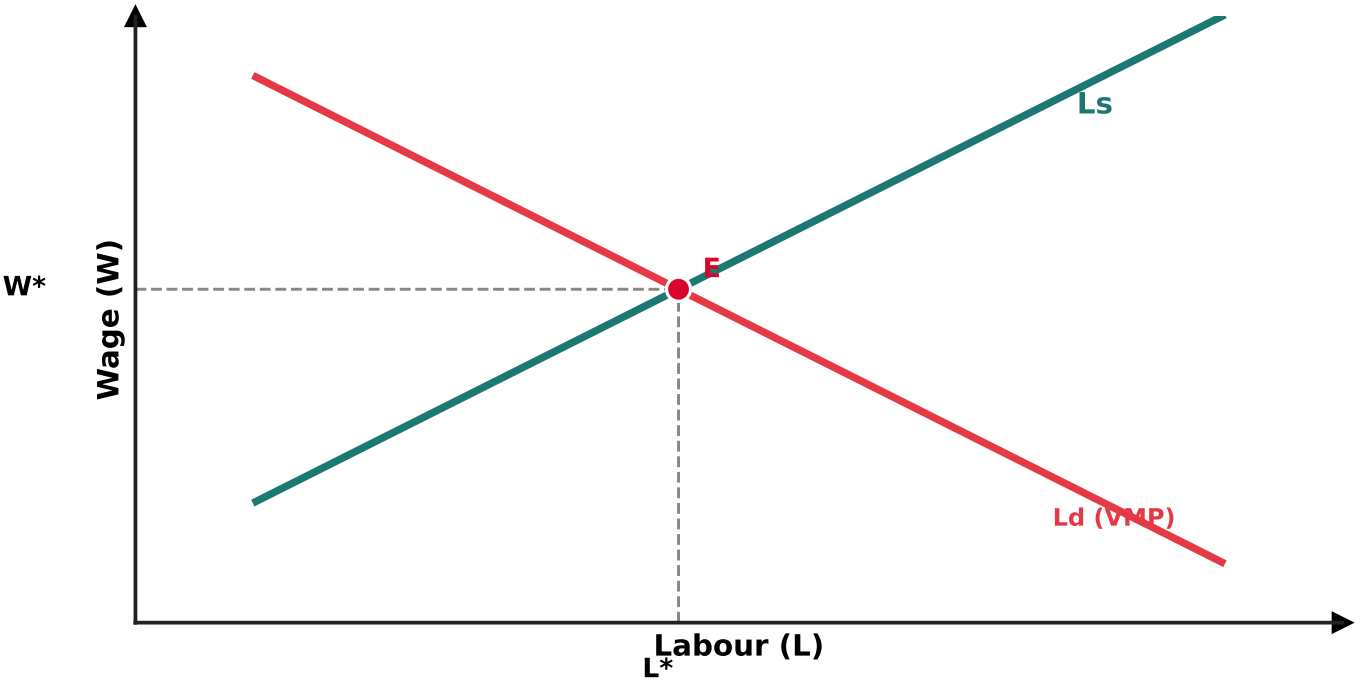
The efficient level of pollution abatement is where Marginal Social Benefit of abatement equals Marginal Social Cost of abatement (E). Beyond E, extra clean-up costs more than the benefit it yields.

54. Environmental Kuznets Curve



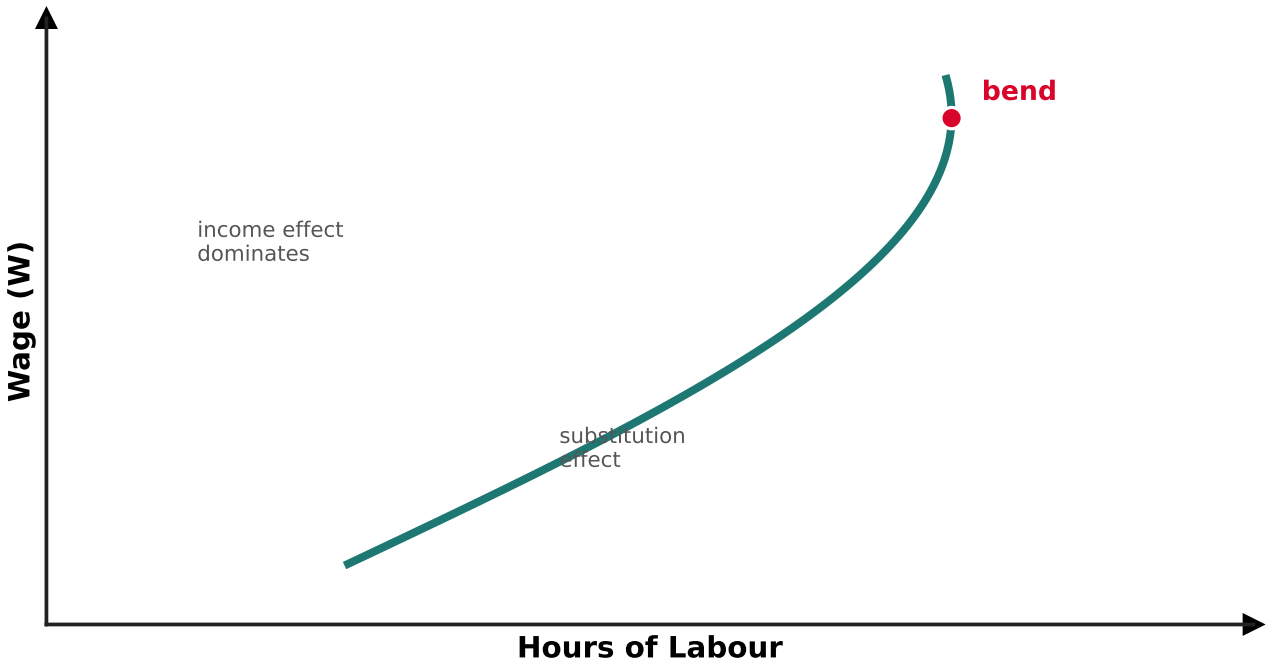
Analogous to Kuznets: environmental degradation rises in early industrialisation, peaks, then declines as richer economies adopt cleaner technology and stricter regulation. Plotted against per-capita income.

55. Labour Demand & Supply



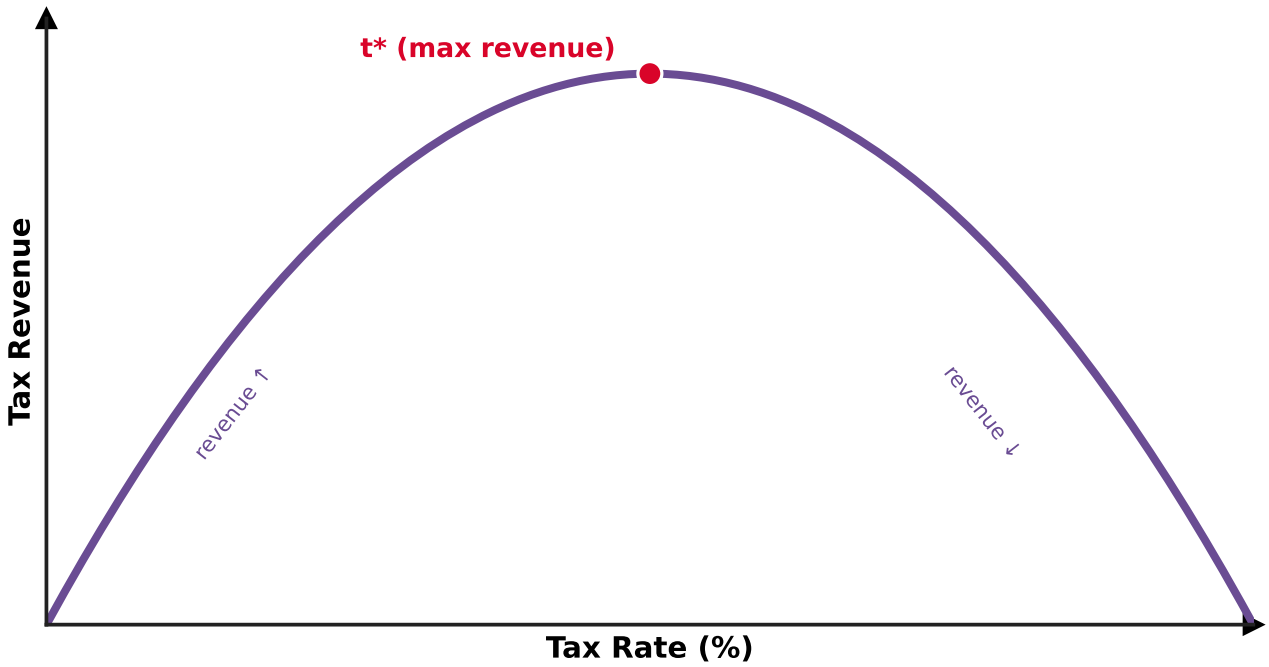
Labour demand (VMP) slopes down; labour supply slopes up. Their intersection sets the equilibrium wage W* and employment L*. Minimum wages above W* create a labour surplus (unemployment).

56. Backward-Bending Labour Supply



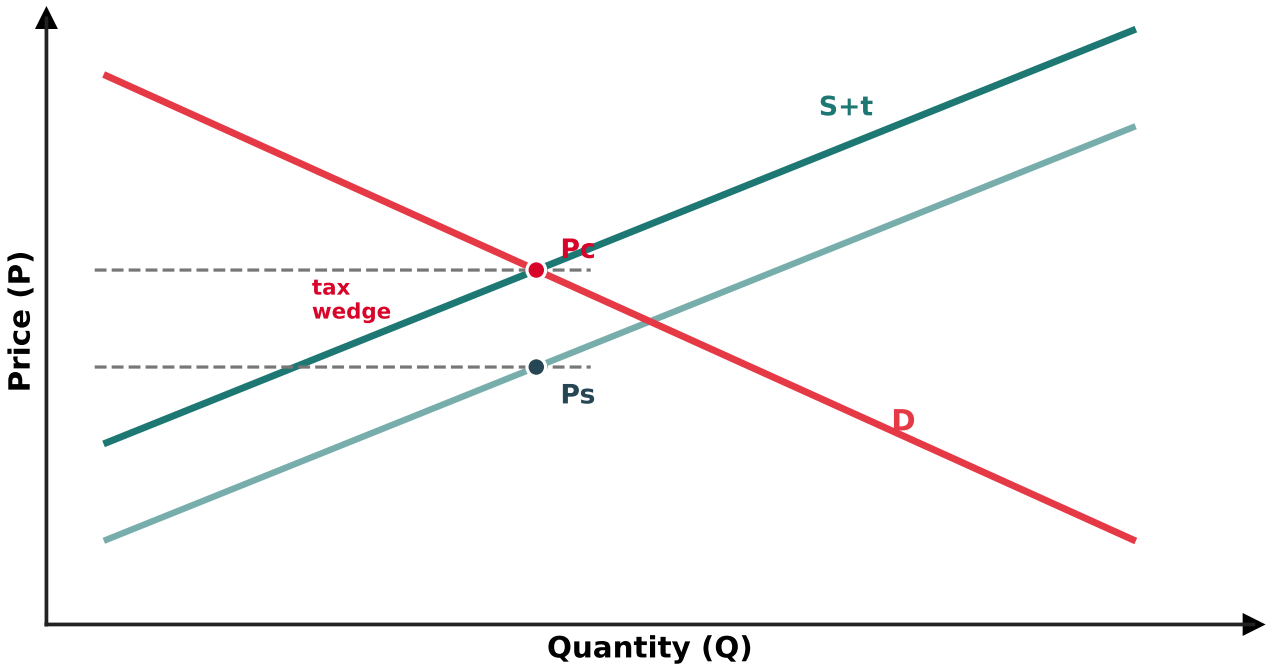
As wages rise, labour supply increases (substitution effect) up to a point; beyond it the income effect dominates and workers choose more leisure, so the supply curve bends backward.

57. Laffer Curve



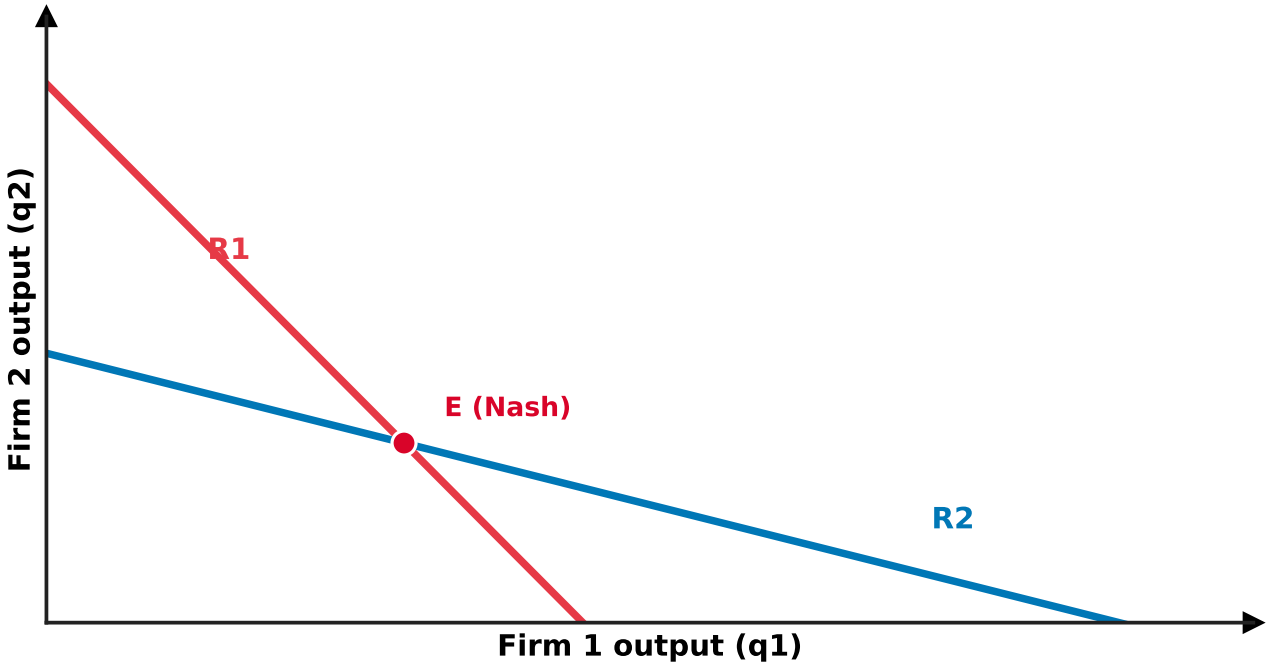
The Laffer curve relates tax revenue to the tax rate. Revenue rises, peaks at the optimal rate t*, then falls as higher rates discourage work and encourage evasion — zero revenue at both 0% and 100%.

58. Tax Incidence (Burden Sharing)



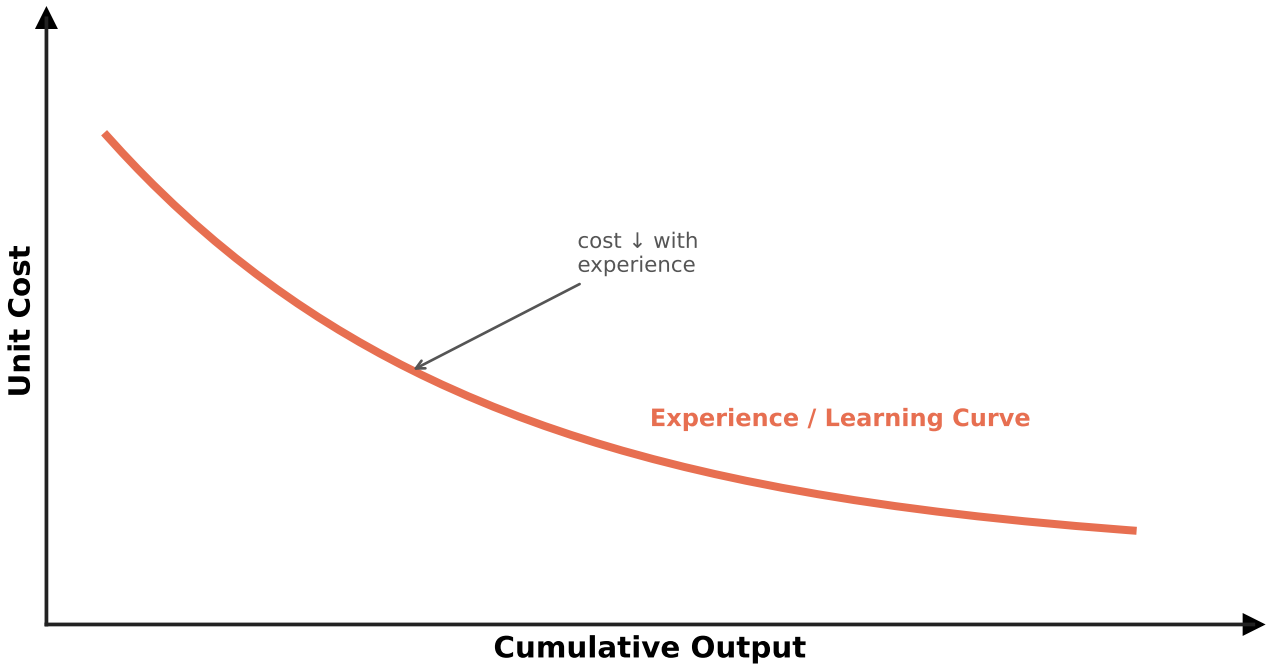
A per-unit tax shifts supply up (S→S+t). The price consumers pay rises to Pc, sellers receive Ps; the gap is the tax. The relative burden depends on demand and supply elasticities; the wedge generates deadweight loss.

59. Cournot Reaction Curves



Each duopolist's reaction (best-response) curve gives its profit-maximising output for the rival's output. The Cournot–Nash equilibrium is where the two reaction curves intersect (E), with mutually consistent quantities.

60. Learning / Experience Curve



As cumulative output grows, unit cost falls because of learning-by-doing and scale efficiencies. The curve is convex and declining — a key source of cost advantage for early/large producers.